

Web Server Security

A

Project Report

Guided by

Prof. Satender kumar

Submitted by

Jandwar Paramveer Singh

Rajinder Singh

Enrolment number:

201918140026



**DEPARTMENT OF
ANIMATION, ITIMS &
MOBILE APPLICATIONS**

Gujarat University

Batch 2022-2024

Gujarat University
Department of Animation,
ITIMS & Mobile Applications



Certificate

Enrolment No: 201918140026

This is to certify that Mr. Jandwar Paramveer Singh student of M.Sc. IT IMS and CS (Integrated) Semester – 8, has duly completed his Term Work for the semester ending in May 2023, in the subject of Project - III towards partial fulfilment of his degree of Master's program.

Date of Submission

13th May 2023

Mentor

Prof. Satender Kumar

ACKNOWLEDGEMENT

It is pleased to take this opportunity to thank all those who helped me directly or indirectly in our research work. I/We would like to express my sincere thanks to internal guide **Prof Satender Kumar**, Department of Animation, ITIMS & Mobile Applications for their valuable help during the work of this thesis. Their suggestions were always there whenever it is needed. His wide knowledge and his logical way of thinking have been great value for me.

Jandwar Paramveer Singh

201918140026

Abstract

The world of communication is every changing and ever growing. We as humans aim to make the process better, faster, and more convenient every single day. But with that drive to achieve convenience we sacrifice other important factors. And one of those factors is security.

With wireless communication the goal has always been to increase convenience. The convenience of connection, convenience of maintenance, convenience of communication and the convenience of evolution. Even though we have had enough experience with this aspect of communication and are in a very good position with it, there are still a number of factors that make it inherently insecure. Which is where this documentation comes in. The entire aim of this document was to focus on what makes a wireless network insecure and what we can do to secure it.

For executing this goal, this document constantly compares different factors of a wireless network with its wired counterpart. Like how its convenience makes it easier to setup, connect, maintain and change but at the same time it makes easier for the attacker to connect, eavesdrop, disrupt and manipulate. This document thus, aims to show to, the inner workings of a wireless network, factors that influence a wireless network, threats and vulnerabilities in a wireless network, possible attacks on a wireless network and the ways to secure a wireless network.

This document keeps a bigger focus on Wi-Fi as a communication than any other form of wireless communication. And it does so because other forms of wireless communications like Bluetooth or broadband have been heavily standardised. To a point that there isn't a huge potential of variance and mistake left. With all of these points in mind, it's time to start his journey of ours.

TABLE OF CONTENTS

<u>Acknowledgement</u>	<u>3</u>
<u>Abstract</u>	<u>4</u>
<u>List of Figures.....</u>	<u>6</u>
<u>List of Tables.....</u>	<u>7</u>
<u>Chapter - 1 How A Wireless Network Works</u>	<u>9</u>
<u>Chapter - 2 What makes a Wireless Network Vulnerable.....</u>	<u>27</u>
<u>Chapter - 3 Wireless Threats and Attacks</u>	<u>37</u>
<u>Chapter - 4 A Secure Wireless Network</u>	<u>58</u>
<u>Conclusion.....</u>	<u>71</u>
<u>References.....</u>	<u>72</u>

LIST OF FIGURES

<u>Figure 1.1 Wired and Wireless Network.....</u>	<u>10</u>
<u>Figure 1.2 Radio Antenna.....</u>	<u>15</u>
<u>Figure 1.3 Modulation.....</u>	<u>19</u>
<u>Figure 1.4 Star Topology.....</u>	<u>23</u>
<u>Figure 1.5 Mesh Topology.....</u>	<u>24</u>
<u>Figure 1.6 Cluster-tree Topology.....</u>	<u>24</u>
<u>Figure 2.1 Cycle of Encryption.....</u>	<u>28</u>
<u>Figure 2.2 WP3 Standard.....</u>	<u>31</u>
<u>Figure 2.3 Bad password saving practices.....</u>	<u>33</u>
<u>Figure 2.4 Disgruntled Employee.....</u>	<u>34</u>
<u>Figure 2.5 Key-card based access control.....</u>	<u>35</u>
<u>Figure 3.1 Rogue AP attack.....</u>	<u>39</u>
<u>Figure 3.2 Denial-of-Service Attack.....</u>	<u>41</u>
<u>Figure 3.3 Wormhole Attack.....</u>	<u>42</u>
<u>Figure 3.4 Sinkhole Attack.....</u>	<u>43</u>
<u>Figure 3.5 Eavesdropping.....</u>	<u>44</u>
<u>Figure 3.6 Wireless ARP Poisoning Attack.....</u>	<u>45</u>
<u>Figure 3.7 Hash Cracking.....</u>	<u>46</u>
<u>Figure 3.8 Side-Channel Attacks.....</u>	<u>48</u>
<u>Figure 3.9 Man trying to access a Wi-Fi Network.....</u>	<u>50</u>
<u>Figure 3.10 Person's activity being spied on.....</u>	<u>51</u>
<u>Figure 3.11 Toaster.....</u>	<u>51</u>
<u>Figure 3.12 Network Diagram of target network.....</u>	<u>52</u>

<u>Figure 3.13 GitHub library for deauther.....</u>	<u>53</u>
<u>Figure 3.14 Code for deauther.....</u>	<u>53</u>
<u>Figure 3.15 Scan of the target network.....</u>	<u>54</u>
<u>Figure 3.16 Beacon attack on the target.....</u>	<u>54</u>
<u>Figure 3.17 Aftermath of beacon attack.....</u>	<u>55</u>
<u>Figure 3.18 De-authentication command execution.....</u>	<u>55</u>
<u>Figure 3.19 Aftermath of de-authentication attack – 1.....</u>	<u>55</u>
<u>Figure 3.20 Aftermath of death attack – 2.....</u>	<u>56</u>
<u>Figure 3.21 Alternative way of de-authentication attack.....</u>	<u>56</u>
<u>Figure 4.1 Layered approach to wireless security.....</u>	<u>59</u>
<u>Figure 4.2 Ranking of encryption algorithms.....</u>	<u>60</u>
<u>Figure 4.3 Wireless WIDS system.....</u>	<u>64</u>
<u>Figure 4.4 Wireless Intrusion Prevention System (WIPS).....</u>	<u>65</u>
<u>Figure 4.5 Wireless Controller device.....</u>	<u>65</u>
<u>Figure 4.6 Wireless VPN Gateway device.....</u>	<u>66</u>
<u>Figure 4.7 Wireless Firewall device.....</u>	<u>66</u>

LIST OF TABLES

Table 1.1.....	25
--------------------------------	--------------------

Chapter - 1

How A Wireless Network Works

Introduction

Before we delve into discussing the threats and possible attacks to a wireless network, we first need to look into how a wireless network actually works. So, in this chapter, we'll look into the need of having a wireless network, types of wireless networks, Workings of a wireless network and a lot more. This chapter aims to provide you with the knowledge required for understanding further chapters and topics.

What is a Wireless Network?

A wireless network allows devices to remain linked to the network yet to roam without any cables unattached. Wi-Fi signals are amplified by access points, meaning a computer can be far from a router but still be linked to the network. You connect to that organization's wireless network when you connect to a Wi-Fi hotspot at a cafe or another public location.

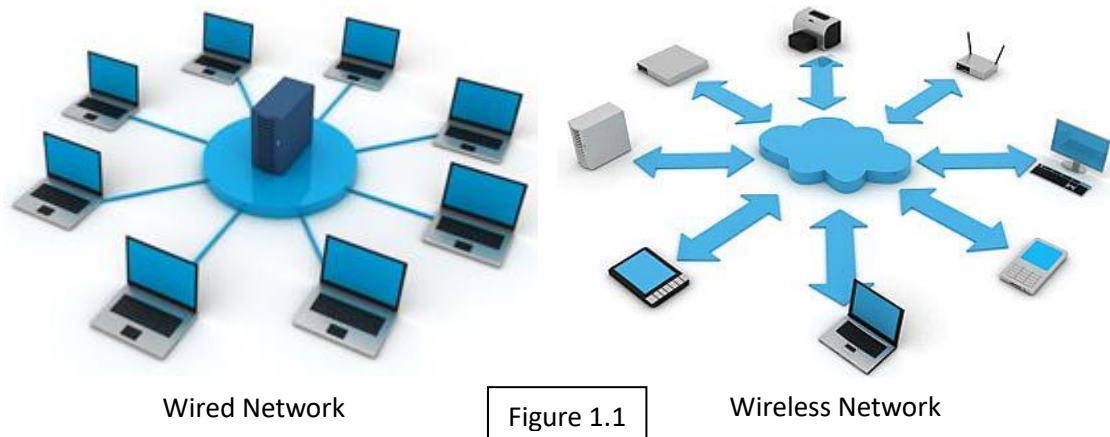
Research into wireless communication was advanced mainly for military use, and various wireless technologies were developed and improved to fulfil that purpose. Wireless communication was brought on a commercial basis in the area of broadcasting, one of the significant applications of wireless communication. Radio broadcasts started in the first half of the 20th century and TV broadcasts started in the second half. Since the late 1980s, wireless communication has been widely used in mobile phones and other mobile terminals by individuals as these technologies, principally in semiconductor and software, have rapidly developed in line with the spread of the new infrastructure of the Internet.

Why Do We Need a Wireless Network?

What makes a wireless network necessary? That's the question here. And there are a few logical answers to this. A lot of it's need came from the problem with the older way of communicating using wires.

- **Limitations with wires:** Having a wired network brings some obvious limitations, like the length of wires in use, the need of drilling holes and changing the physical structure

of buildings to set up wired communication, and the possible physical damage to said wires. Such factors don't affect wireless communication.



- **Cost of wires:** Setting a big network, of connecting many systems together requires a bunch of wires, and that too most probably highspeed wires. Such a setup isn't cheap. Although wireless communication is not faster than wired one, its setup requires only a single or a few access points instead of many wires. And that convenience itself makes up for lesser speeds in many ways. Such a choice is made based on what is required.
- **Complexity of setting up wired communication:** While setting up a wired connection, you have to think of the number of clients, their position, required throughputs, the topology, network devices and more. Such factors showcase the obvious complexity behind creating wired networks.
- **Knowledge required to pick and use the right wires:** Knowing which wires to pick for your infrastructure based on length, throughput and topology requires knowledge and experience. There is no doubt that picking the right router, ISP and package requires knowledge too, but the complexity is decreased in the case of wireless networks.

Advantages of Having a wireless network

- Freedom from wires: Our discussion above proves how advantageous that is
- Easy to setup: Wireless network is relatively easy to setup and then expand.
- Better or global coverage: Many rural areas or remote areas are connected in such way as setting up a number of access points can get the job done.
- Flexibility: Wireless network is more flexible and adaptable when compared to a wired network.
- Cost-effectiveness: As we discussed earlier how costly wired networks can be, the wireless network proves to be relatively cheaper.

- Mobile and portable: Wireless network is easy to carry and re-install in another place.
- Mobility -: It has good mobility of usage.
- Data transmission is fast: In wireless network data transmission is fast. Not always faster than wired, but faster than most of the low speed wired connections that companies might use.
- Low maintenance: In any wireless communication low maintenance cost. You don't have to be worried about replacing damaged wires or anything like with wired networks.
- It can be accessed from anywhere and anytime. for ex. professional can complete their work from remote location.
- Network planning: Network planning in wireless communication is very easy as compare to wired network. Wireless software configuration of frequency, power and other parameter get the job done. While a wired connection requires you to set up entire topologies.
- Expandable: Wireless networks can serve a large number of clients without resorting the new hardware. You can add repeaters, amplifiers and multiple access point to expand a wireless network's usability

A few disadvantages of a wireless network

- Lack of security due to the communication being open
- Unreliable due to it being open to interference and chance of jamming.
- Transmission speed is comparably less
- it has a limited amount of bandwidth for communication and breaches of network security.
- Wireless networks can be easily hacked.
- Wireless networks require a careful radio frequency when they are installed, not being careful can cause breaking of laws, like accidentally using frequencies registered for military use
- Wireless networks are usually inexpensive, but the cost of installation is very high, setting up a wireless network on a small scale is fine, but on a large scale it can get very costly
- Still requires experience from the person setting it up

Types of Wireless Networks

- 1) **WPAN:** Also known as piconet. A small-scale wireless network that requires little or no infrastructure and operates within a short range. A WPAN is typically used by a few devices in a single room instead of connecting the devices with cables. Bluetooth is a technology used in this type, wireless headphones, mouse etc, are examples of that.
 - The range is short—about 20 feet
 - Eight active devices
 - Unlicensed 2.4-GHz spectrum
- 2) **WLAN:** WLANs are designed for a larger area than that of a WPAN. These can scale from very small home offices to large enterprise networks. The fact that they are local-area means that the organization where the WLAN exists also manages and probably owns the equipment
 - A larger range than a WPAN—close to 100 meters from AP to client
 - WLANs are very flexible, so more than eight active devices/clients are expected, unlike a WPAN
 - 2.4-GHz or 5-GHz spectrum
- 3) **WMAN:** A wireless metropolitan-area network (WMAN) covers a large geographic area. A WMAN is mostly governed by a single entity such as an Internet Service Provider (ISP), government entity, or any other large corporation. The user has to take the authorized access from the providers to use WMAN as the access is restricted only to the authorized users/subscribers. It's most well known as WiMAX
 - Speeds decrease as the distance increases.
 - Close to broadband speeds versus Ethernet speeds.
 - Used as a backbone, point-to-point, or point-to-multipoint.
- 4) **WWAN:** A wireless wide-area network (WWAN) covers a large geographic area. It provides regional, nationwide and global wireless coverage. WLAN (Wireless Local Area Network) differs from WWAN (Wireless Wide Area Network) technology wise for example when WLAN uses Wi-Fi to connect and transfer data, WWAN uses telecommunication cellular network technologies such as 2G, 3G, 4G LTE, and 5G to transfer data.
 - Low data rates
 - Pay-for-use
 - High cost of deployment

Types Of Wireless Communication

Before we get into types of wireless communication, we'll have to look into the classification that all kinds of networks are put into.

A wireless Communication system is classified into Simplex, Half Duplex & Full Duplex. The simple wireless communication system is one-way communication. In this type, the communication can be done in one direction only. The best example is the radio broadcast system. The half-duplex communication system is two-way communication; however, it is not simultaneous. The best example of this type of communication is walkie – talkie.

The full Duplex communication system is also two-way communication & it is simultaneous. The best example of this communication system is the mobile phone. In wireless communication, the devices which are used for communication may change from one service to others because these are available in different shape, size & data throughput. Let's now look into different types of wireless communication.

- 1) **Satellite communication:** Satellite communication is one type of self-contained wireless communication technology; it is widely spread all over the world to allow users to stay connected almost anywhere on the earth. When the signal (a beam of modulated microwave) is sent near the satellite then, the satellite amplifies the signal and sent it back to the antenna receiver which is located on the surface of the earth. Satellite communication contains two main components like the space segment and the ground segment. The ground segment consists of fixed or mobile transmission, reception, and ancillary equipment and the space segment, which mainly is the satellite itself.
 - i) **GPS:** This kind of system is used to help by providing different wireless services such as speed, location, navigation, positioning using satellites, and GPS receivers.
- 2) **Infrared Communication:** Infrared wireless communication communicates information in a device or system through IR radiation. IR is electromagnetic energy at a wavelength that is longer than that of red light. It is used for security control, TV remote control, and short-range communications. In the electromagnetic spectrum, IR radiation lies between microwaves and visible light. So, they can be used as a source of communication. For successful infrared communication, a photo LED transmitter and a photodiode receptor are required. The LED transmitter transmits the IR signal in the form of nonvisible light, that is captured and saved by the photoreceptor.

- 3) **Broadcast Radio:** Radio waves are electromagnetic signals, that are transmitted by an antenna. These waves have completely different frequency segments, and you will be ready to obtain an audio signal by changing into a frequency segment. Handy multichannel radios permit a user to speak over short distances, whereas citizen's band and maritime radios offer communication services for sailors.

For example, you can take a radio station. When the RJ says you are listening to 92.7 BIG FM, what he really means is that signals are being broadcasted at a frequency of 92.7megahertz, which successively means the transmitter at the station is periodic at a frequency of 92.700,000 cycles/second.

- 4) **Microwave communication:** Microwave wireless communication is an effective type of communication, mainly this transmission uses radio waves, and the wavelengths of radio waves are measured in centimetres. In this communication, the data or information can be transfers using two methods. One is the satellite method and another one is a terrestrial method.

Wherein satellite method, the data can be transmitted through a satellite, that orbit 22,300 miles above the earth. Stations on the earth send and receive data signals from the satellite with a frequency ranging from 11GHz-14GHz and with a transmission speed of 1Mbps to 10Mbps. In the terrestrial method, in which two microwave towers with a clear line of sight between them are used, ensuring no obstacles to disrupt the line of sight. The frequency range of the terrestrial system is typically 4GHz-6GHz and with a transmission speed is usually 1Mbps to 10Mbps. The main disadvantage of microwave signals is, they can be affected by bad weather, especially rain.

- 5) **Wi-Fi:** Wi-Fi is a low power wireless communication, that is used by various electronic devices like smartphones, laptops, etc. In this setup, a router works as a communication hub wirelessly. These networks allow users to connect only within close proximity to a router. Wi-Fi is very common in networking applications which affords portability wirelessly. These networks need to be protected with passwords for the purpose of security, otherwise, it will access by others.

In fact, most of the conversations that we have in this document will be on Wi-Fi networks

- 6) **Mobile Communication Systems:** The burgeoning mobile phone industry uses similar technology to Wi-Fi but on a much grander and safer scale. Mobile phone companies provide coverage to customers nationwide or even international scale.

They do this by means of a complex blend of local networks and transmitters together with satellite support.

- 7) **Bluetooth Technology:** The main function of Bluetooth technology is that permits you to connect various electronic devices wirelessly to a system for the transferring of data. Cell phones are connected to hands-free earphones, mouse, wireless keyboard. By using the Bluetooth device, the information from one device to another device. This technology has various functions and it is used commonly in the wireless communication market.

8) **Other forms of Wireless Communication**

- **Radar:** Radar is an electromagnetic sensor or detection system used to track, locate, detect & and identify objects of different types at significant distances. The operation of this detection system can be done by sending electromagnetic energy in the direction of objects, usually called targets, which observes the echoes. The targets may be ships, astronomical bodies, aircraft, spacecraft etc.
- **Radio Frequency Identification:** Radio Frequency Identification (RFID) is one kind of wireless communication that uses electrostatic coupling otherwise electromagnetic in the RF portion of the electromagnetic spectrum to exclusively identify a person, object & animal. RFID & barcode technology are used in related methods to track inventory like in manufacturing, healthcare, shipping, etc.

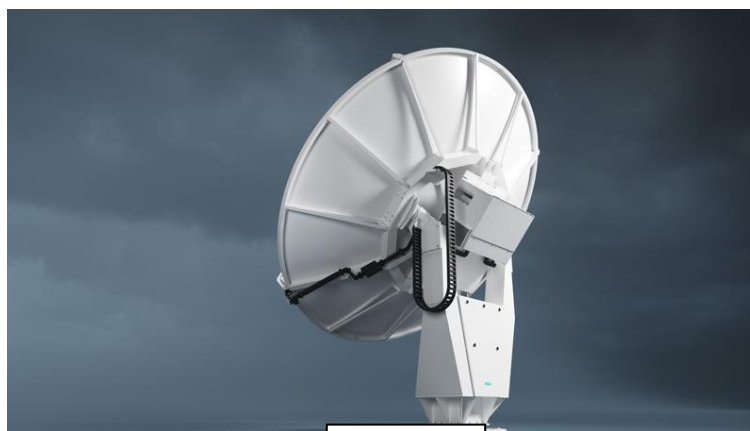


Figure 1.2

Factors affecting Wireless Networks

Wireless signals can have trouble penetrating solid objects which can be any numbers of things such as hills, buildings, single walls or even people. The more obstructions you have between the transmitter and receiver, the more chance there is that the signal strength will be affected so you should try and maintain as clearer line of site as possible. Not being careful about the following factors will sabotage your network setup plans.

Network Range & Distance Between Devices

The further apart the networked devices that are trying to communicate with each other are, the more the signal strength drops. This is due to way that wireless signals propagate covering a wider area as they travel further and because of this, as the signal spreads more, the weaker it becomes. The signal strength decreases, roughly, in an inverse cubic relation with respect to the distance between the two devices.

Wireless Network Interference

Wireless Networking is becoming more and more common and therefore more wireless transmissions are being sent through the air. Signals operating at similar frequencies can cause interference with each other and have a significantly negative effect on the performance of the network. This means that more popularly used frequency bands such as the 2.4GHz unlicensed band can get severely affected by the overcrowding of wireless signals to a point where a device will not operate at an acceptable level.

Signal Sharing

Wireless Networks allow more than 1 person to communicate with another network source at any one time. This sharing of connection means that the more subscribers utilising the network, the more devices the access point has to try and communicate with instantaneously. The point of access has to delegate its resources to each subscriber individually per the number of transmitting radios it operates on.

Network Usage & Load

You will find that the more clients that are utilising the network bandwidth, the less that there is to share between them. Higher throughput speeds are then required to keep the network reliable.

Poorly Deployed Antennas

As antennas work by limiting the spread of transmitted signals to a specific direction or orientation, they must be installed and mounted correctly to improve network performance. The most obvious is the highly directional antenna type that must be pointed directly at the receiver in order for the signal to be picked up but other types of antennae must be orientated correctly to ensure that they are transmitting signal on the correct plane.

Local Environment Characteristics

Wire meshing within the walls can cripple signal transmissions and the materials used have different levels of effect; concrete is a usual suspect in badly performing indoor networks. It almost goes without saying that the thicker the walls, the less success the signal will have penetrating through it whilst maintaining a high strength.

Spectrum Channel limitations

Certain bands like the 2.4GHz provide only three proper channels for communication, 1,6 and 11. And they do this to avoid overlap in between the total of 11 channels that it has. These channel limitation in a spectrum serves as a pretty influential factor affecting a wireless network.

Signal Reflection

Reflection happens when a signal bounces off of something and travels in a different direction. This can be illustrated by shining a flashlight on an angle at a mirror, which causes it to reflect on an opposite wall. The same concept is true with wireless waveforms. Although offices do not usually have mirrors lying around, they do have other objects with similar reflective qualities, such as monitors and framed artwork with glass facing

Wireless Signal Restriction

For security reasons you may want to restrict the transmission of your wireless signal to only the areas that you want to provide network access for. This can be troublesome as controlling signal propagation is not easy as it can travel through walls into other buildings or outside areas where someone could try to connect without authorisation.

Other lesser-known factors are,

- **Transmitter Power limitations**

- **Backwards compatibility with older standards** i.e., newer protocols have to limit themselves if one of the devices use an older protocol for connecting to an access point
- **Polarisation of Signal** i.e., Antennas polarise their transmitted signals and therefore receiving devices should be orientated to receive the signal on the same plane for the optimum performance.
- **Speed loss due to Wireless Overheads** i.e., due to encryption, packet translation and partial utilisation of channel bandwidth for user data, advertised throughput speeds are not the actual, real world throughput speeds
- **Lowering of performance by devices to stay connected to the network**

Important Functions and Factors affecting a wireless network

Modulation

The process by which data/information is converted into electrical/digital signals for transferring that signal over a medium is called modulation. It increases strength for maximum reach of the signals. The process of extracting information/data from the transmitted signal is called demodulation. A Modem is a device that performs both modulation and demodulation processes.

Now why do we need to modulate signals? Well, modulation allows us to send a signal over a bandpass frequency range. If every signal gets its own frequency range, then we can transmit multiple signals simultaneously over a single channel, all using different frequency ranges. Another reason to modulate a signal is to allow the use of a smaller antenna.

When it comes to modulation types, there are multiple, we are going to focus on just one for understanding purposes, OFDM. Orthogonal Frequency Division Multiplexing (OFDM) is another widely used modulation method used to achieve high data rates and spectral efficiency. It is known as a multicarrier modulation method as many carriers are used instead of just one. What this means is that this will allow multiple signals to travel in the same frequency band without overlapping with each other. Thus, allowing multiple smooth communications in the same area.

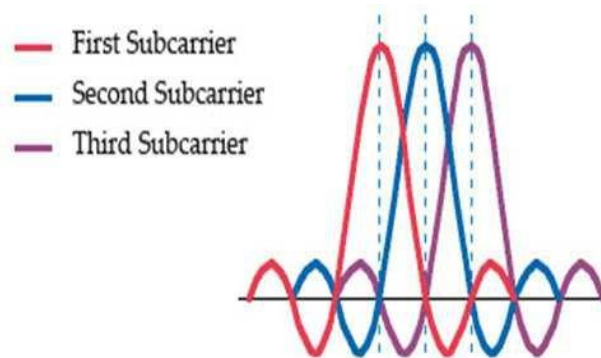


Figure 1.3

Frequency regulation

Frequencies are divided for required purposes. For example, in India 2.4 GHz-2.4835 GHz are used for Indoor Wi-Fis or low power wireless equipment. These regulations are put in place for optimal management and to avoid as much overlap as possible. Higher frequencies by default perform better outdoors and thus are registered for such. Plus, in every country there is a set of frequency that is designated only for military communications. Here is a short table of how frequencies are registered in India.

2.4-2.4835 GHz	- Indoor - Low power wireless equipment – Wi-Fi
2.4-2.4835 GHz	- Low power wireless equipment - Zigbee
5 GHz	- Indoor - Low power wireless equipment
5.825-5.875 GHz	- Outdoor
865-868 MHz	- Outdoor - RFID - LoraWAN - 865 to 867 MHz
402-405 MHz	- Medical Usage

Beacons and Probes

Beacon frame is one of the management frames in IEEE 802.11 based WLANs. It contains all the information about the network. Beacon frames are transmitted periodically, they serve to announce the presence of a wireless LAN and to synchronise the members of the service set. To connect to a Wi-Fi hotspot, devices send out a probe request to available network access points (APs) in their surroundings. An available network sends a probe response, initiating the connection. While this process is standard, these requests also serve as a means to track, trilaterate [locate], and identify devices for attackers who passively sniff network traffic.

SSIDs

An SSID is a name or number a device sees whenever it searches for a network. It can belong to a business or home network and can have a default name, generally the name of network

carrier or router manufacturer, or a custom name created by the owner. SSIDs prove as a way to recognise a network. Thus, helping securing the network too, an attacker won't be able to connect to a wireless network without knowing the network's SSID. SSID can be hidden, only allowing a new client to connect if they know the rest of the details about the network. Another big purpose of SSID comes in roaming, if you want multiple access points in your network to allow roaming then all those access points need to have the same SSID.

CSMA/CA

Carrier Sense Multiple Access - Collision Avoidance (CSMA/CA). The aim of CSMA/CA is straight up to avoid collisions in wireless networks.

With wired networks there is both CSMA/CD and CSMA/CA where CD refers to collision detection and CA refers to collision avoidance. With a wired network it is possible to detect a collision by observing the change in amplitude of the signal in the wire, but something like this is not possible in a wireless network, you cannot just measure the air can you. But what you can do that in wireless network, the best you can do here is straight up avoid a collision. Thus, CSMA/CA. In collision avoidance the sender and receiver maintain a mutual communication where they let each other know if network is available for data transfer, they do this by using RTS, CTS and ACK. Where RTS is request to send and CTS is clear to send.

RTS is sent by a device that wants to send the message, it asks other devices in the network if they can send their data. CTS is sent by other devices, signifying the device who sent that RTS that they are clear to send their data. ACK helps the connection by acknowledging receipt of frames.

IFS

IFS is a period that a station has to wait before it can send. Not only does IFS ensure that the medium is clear, but it ensures that frames are not sent so close together that they are misinterpreted.

Chipping Codes

To handle the possibility of noise interference with a wireless transmission, DSSS uses a sequence of chips. When DSSS spreads information across a frequency range, it sends a single data bit as a string of chips or a chip stream. With redundant data being sent, if some of the signal is lost to noise, the data can likely still be understood. The chipping code process takes each data bit and then expands it into a string of bits.

The chipping code for the bit value of 1 is expanded to the chip sequence of 00110011011, and the chipping code for the bit value of 0 is 11001100100. Therefore, after the data bits are sent, 1001 creates the chip sequence of: -

00110011011-11001100100-11001100100-00110011011

In this way, even if some of the bits are lost in transmission, the receiver can identify the ones and zeros.

Dynamic Rate Shifting (DRS)

This refers to changing the speed of transmission in a wireless network to avoid loss of data. One of the factors that make this feature necessary is distance from an access point. In 802.11 networks, operating in the 2.4-GHz range, the devices can rate-shift from 11 Mbps to 5.5 Mbps, and further to 2 and 1 Mbps depending on the circumstances. It even happens without dropping your connection. Also, it is done on a transmission-by-transmission basis, so if you shift from 11 Mbps to 5.5 Mbps for one transmission and then move closer to the AP, it can shift back up to 11 Mbps for the next transmission.

Link Budget

A link budget is an accounting of all of the power gains and losses that a communication signal experiences in a telecommunication system. By knowing all the losses and gain in our communication system, we can predict how reliably the signal can be received. For example, in a satellite communication link the provider would need to know the satellite transmitter power output, the gain provided by the transmit antenna, propagation loss etc. in order that the receiver can still receive the signal under adequate signal to noise ratio. Link budgets are useful in point-to-point communication where the link can be optimised with the use of highly directional transmitting and receiving antennas.

Polarization

The polarization of an antenna is loosely defined as the direction of the electromagnetic fields produced by the antenna as energy radiates away from it. These directional fields determine the direction in which the energy moves away from or is received by an antenna. This polarization can be of three types, vertical, horizontal, and circular. Right type of polarization is decided based on the use case, like where the antenna is going to be used, for what range and for what type of coverage. Based on that different antennas are made for different purposes, like for coverage on a single floor, in a single direction, through multiple floors etc.

Diversity

This is made to fight the multipath issue. This issue arises when signals are reflected through the environment causing the receiver to receive the same signal from multiple directions. One way to deal with multipath issues is to use two antennas on one AP. Diversity is the use of two antennas for each radio to increase the odds that you receive a better signal on either of the antenna.

Wireless Network Devices

Wireless Router



This device acts as a medium between the internet connection and other wireless devices. It accepts incoming internet connections and then sends the data in radio frequency signals to nearby wireless devices. The role of a wireless router is to connect devices like computers and mobiles to the internet.

Wireless Bridge

Wireless bridges are generally required to provide links over long distances. For example, when you have two or more LANs within a few miles of each other and you want to link them, you will use a wireless bridge.

Wireless Repeater

This networking device can help extend the range of a router. When you place it between the computer connected to the internet and a wireless router, it boosts the signal strength. The role of a repeater is to get the wireless signals and then re-emit them after increasing their power. As a result, you get faster connection speeds.

Wireless Adapters

These are the devices in your system that allow you to connect and communicate in a wireless network. Nowadays, most computers have wireless adapters in their motherboard. Still, if it's not available on your computer, then you can install stand-alone wireless adapters. It will ensure that you can connect your computers to the internet.

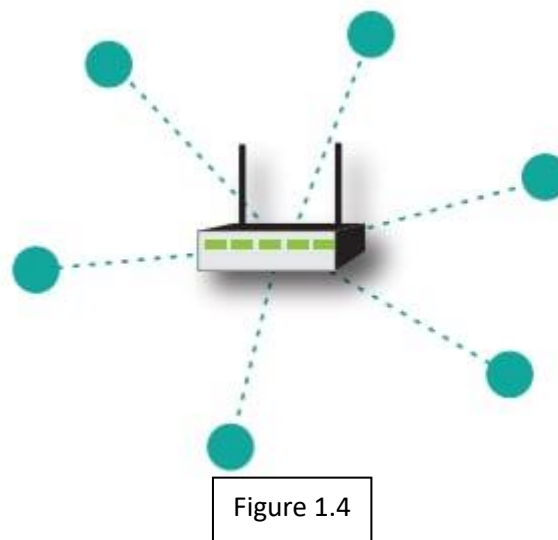
Wireless Network Topologies

Star Topology

A star topology is a point-to-point architecture where individual wireless devices, or nodes, communicate directly with a central "hub" or gateway.

The gateway then transmits the data to a central collection point, such as a control room – either directly, or by connecting to another network. The gateway also receives data from the central point and relays it to the appropriate device. Because each device communicates directly with the gateway, a star topology is also sometimes described as a "point-to-point" or "line-of-sight" architecture.

These simple, direct wireless connections make star topologies applicable for relatively simple or low-power applications because they can potentially use the least amount of power of all the topologies.



Mesh Topology

Unlike in a star topology, where each device talks only to the gateway, the devices in a mesh topology can also communicate with other nodes in the network. A message can "hop" from node to node to node until it reaches the assigned gateway – a capability called multi-hopping.

This can have it's own advantage like Easier expansion and high fault tolerance, but will also potentially consume more power.

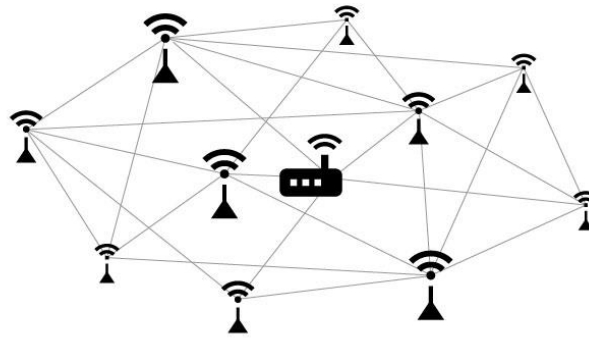


Figure 1.5

Cluster-tree topology

A cluster-tree topology is a hybrid of star and mesh topologies: Wireless devices in a star (point-to-point) topology are clustered around routers or repeaters that communicate with each other and the gateway in a mesh (point-to-multipoint) topology.

A cluster-tree topology combines point-to-point communications from device to router with point-to-multipoint communications between routers and gateways.

This approach attempts to blend the advantages of both other topologies: potentially low power consumption of the "star" portions of the cluster-tree, and extended range and fault tolerance of the "mesh" portions.

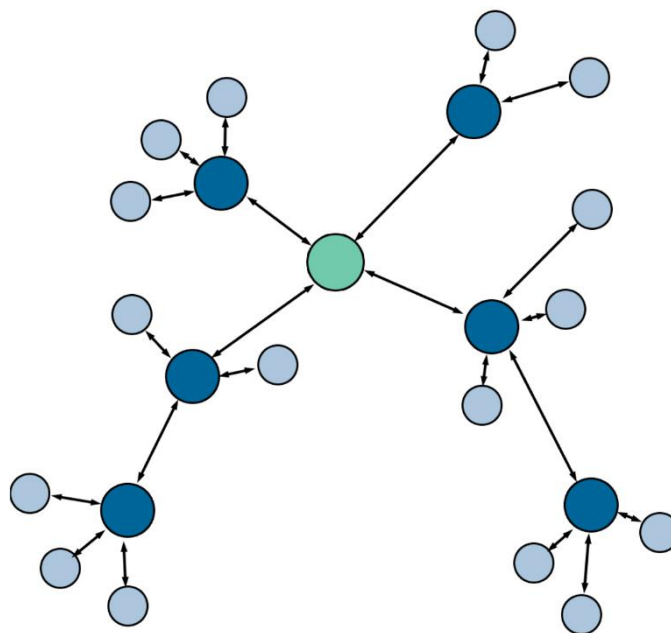


Figure 1.6

Original 802.11 Topologies

- **Ad-hoc networks:** When two computers want to communicate directly with one another, they do so in the form of an ad hoc network. Ad hoc networks don't require a central device to allow them to communicate. Rather, one device sets a group name and radio parameters, and the other uses it to connect. This is called a Basic Service Set (BSS), which defines the area in which a device is reachable.
- **Infrastructure mode network:** Infrastructure mode is an 802.11 networking framework in which devices communicate with each other by first going through an Access Point (AP). Star topology would be the optimal example of this.

Antennas and it's types

Antennas as we know are used for sending and capturing Radio frequencies (RF). And it has its own types, each made to serve a specific purpose.

Omnidirectional: These antennas radiate radio frequency (RF) in all directions. They prove to be very useful in home or office settings where the user aims to set up a router and forget about the rest.

Semi-directional: These antennas point the RF signal in a specific direction for point-to-point communication. Can be used for short to medium distance communication, indoors or outdoors. These are used in a campus like environment for communication between two buildings.

Highly directional: These are highly focused and high gain antennas with a narrow beam width. These antennas are used for long distance point-to-point communication. A TV antenna dish is a great example of this.

Wireless Protocols

Protocol	Frequency	Bandwidth	Data Rate Per Stream
802.11n	2.4 GHz - 5 GHz	20,40 MHz	100 MBPS or more
802.11g	2.4 GHz	20 MHz	Up to 54 MBPS
802.11b	2.4 GHz	20 MHz	Up to 11 MBPS
802.11a	3.7,5 GHz	20 MHz	Up to 54 MBPS
802.11ac	5 GHz	20,40,60,80MHz	Up to 720 MBPS

Table 1.1

Summary

This chapter was all about getting the right knowledge for what's to come next. In this chapter we talked about why a wireless network is even needed. Then We looked into the types of wireless networks and how different wireless communications function. Then we discussed the factors that affect a wireless network and the techniques used to deal with different possible issues in a wireless network. After that we looked at different network devices, topologies and antennas. And finally, we ended by having a brief look at all the IEEE 802.11 protocols, their speeds, bandwidths and frequencies.

Chapter - 2

What makes a Wireless Network Vulnerable

Introduction

With having understood the workings of a wireless network, it is now time to understand what makes a wireless network hackable. In more specific terms, what is it that makes a wireless network vulnerable. Truth be told most of the security for a wireless network depends upon the encryption used for communication. But what adds depth to this topic are the factors that make a wireless network accessible in the first place. For example, even the strongest encryption will do no good if the users don't handle the passwords correctly. So, without further a due, let's look into factors and techniques that make a wireless network vulnerable.

Authentication based problems

When it comes to authentication, there are two authentication modes;

- Open System Authentication Process: It by default provides access without any authentication, but can be used alongside WEP for authentication. It only requires for the end device to be aware of the SSID of the network. As long as the SSID is known, the device will be allowed on to the network. Problem with this system is the SSID is typically broadcasted, and if not, it can be easy to figure out with passive capturing techniques.
- Shared authentication method is commonly used to individual and small business wireless LAN implementations. The method uses a shared-key that is given on both sides of the connection; if the key matches, then the device is allowed onto the network.
- Thirdly, Extensible Authentication Protocol. It is the most commonly used in enterprises as it utilizes an authentication server that is queried for authentication using a variety of credential options. Making this the most secure option of all.

Out of these three methods, the most secure option can be easily pointed out. But even the right authentication won't do anything if the connection isn't encrypted. So, let's have a look at the problems that come with encryption and its implementation. Now to discuss these problems we first need to understand encryption.

Encryption

Encryption is the method by which information is converted into secret code that hides the information's true meaning. The science of encrypting and decrypting information is called cryptography.

In computing, unencrypted data is also known as plaintext, and encrypted data is called ciphertext. The formulas used to encode and decode messages are called encryption algorithms, or ciphers.

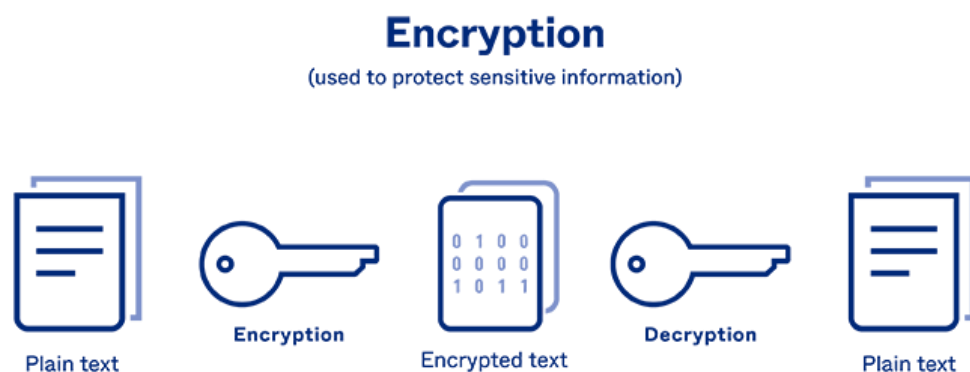


Figure 2.1

To be effective, a cipher includes a variable as part of the algorithm. The variable, which is called a key, is what makes a cipher's output unique. When an encrypted message is intercepted by an unauthorized entity, the intruder has to guess which cipher the sender used to encrypt the message, as well as what keys were used as variables. The time and difficulty of guessing this information is what makes encryption such a valuable security tool.

Encryption based problems

Encryption is an integral part of wireless security. And there are many types of encryptions that can be adopted in a wireless network. Now, admittedly the conversation is not that interesting with Bluetooth. Yes, Bluetooth does use encryption, Bluetooth 5.0 uses AES-CCM encryption but that is pretty much where the conversation ends. Bluetooth is used in very limited terms to everyone tends to simply use the latest version and doesn't thing about the rest. Thus, our conversation will mainly be based around Wi-Fi encryption.

WEP

Wired Equivalent Privacy (WEP) was the first attempt at wireless protection. The aim was to add security to wireless networks by encrypting data. This is accomplished by encrypting data

with the symmetric Rives Cipher (RC4) encryption algorithm, which is a cryptographic mechanism used to defend against threats. WEP is a stream cipher that uses RC4 to produce a stream of bytes that are XORed with plaintext. You don't need to understand how this algorithm works in detail to know what makes it insecure. Here are its flaws that make it vulnerable;

- No defined method for encryption key distribution: Keys shared under this encryption are rarely (if ever) changed. Plus, it is easy to the number of plain text messages encrypted with the same key.
- RC4 was designed for more randomized environments not for simple encryption algorithms like WEP: This again comes back to the key ever being changed. And if the attacker due to this lack of change, figures out the cipher text and that cipher's plain text, then the attacker can compute the actual key. Making all communications open to understand for the hacker.
- Passive data capture techniques can be used by attackers to gather data and crack captured WEP keys.

The algorithm did function well at first, but as computing power increased, the process of breaking encryption became faster. Thus, rendering this algorithm useless in today's world.

WPA

Introduced in 2003, this protocol was the Wi-Fi Alliance's replacement for WEP. It shared similarities with WEP but offered improvements in how it handled security keys and the way users are authorized. While WEP provides each authorized system with the same key, WPA uses the temporal key integrity protocol (TKIP), which dynamically changes the key that systems use. This prevents intruders from creating their own encryption key to match the one used by the secure network. The TKIP encryption standard was later superseded by the Advanced Encryption Standard (AES), but this happened later in its life time after WPA 2 came out. Even with these positives, this algorithm too was eventually found to be insecure:

- WPA PSK (pre-shared key) was found to be vulnerable to cracking attacks. And If PSK is captured, attackers could decrypt all the packets encrypted with that key.
- It was vulnerable to packets spoofing/injection attacks
- The temporal key we talked about was predictable
- TKIP vulnerabilities allow attackers to guess the IP address of the subnet and inject small packets into the network to downgrade the network performance

WPA 2

WPA 2 came with AES (Advanced Encryption Standard) which is a strong encryption algorithm. And it uses a block chaining method known as CCMP (Counter Mode Cipher Block Chaining Message Authentication Code Protocol) It provides stronger data protection and network access control than WPA. Furthermore, it gives a high level of security to the Wi-Fi connections so that only the authorized users can access the network. It came with two modes, one being Personal and other being Enterprise. Where personal mode used the pre-shared key (PSK) method and Enterprise mode used EAP or RADIUS for centralized server authentication. Even with all of this going for the algorithm, it was eventually found to have a few problems too:

- WPA 2 pre-shared key (PSK) was also seen to be vulnerable to attacks like eavesdropping, dictionary and password cracking attacks
- the hoel96 vulnerability in the WAP2 allows attackers to exploit a shared group temporal key to perform MITM (Man-in-the-Middle) attack and DoS attack.
- The temporal key was still predictable due to the use of an insecure Random Number Generator (RNG).
- The KRACH exploit in WPA2 may allow attackers to sniff packets, hijack connections, inject malware, and decrypt packets.
- Attackers can exploit the WPA2 replay attack detection feature to send forged group-addresses data frames with a large PN to perform a DoS attack.

WPA 3

Then came WPA 3. Introduced in 2018, WPA 3 aimed to resolve all the security concerns in past versions and it introduced new features for personal and enterprise use, some of it are.

- Individualized data encryption: When logging on to a public network, WPA3 signs up a new device through a process different from shared password. WPA3 uses a Wi-Fi Device Provisioning Protocol (DPP) system that allows users to use Near Field Communication (NFC) tags or QR codes to allow devices on the network. In addition, WPA3 security uses **GCMP-256 encryption** rather than the previously used 128-bit encryption. A bigger key means a much longer time for anyone to be able to decrypt it.
- WPA3 protects against offline password guesses by allowing a user only one guess, forcing the user to interact with the Wi-Fi device directly, meaning they would have to be physically present every time they want to guess the password. WPA2 lacks built-in

encryption and privacy in public open networks, making brute force attacks a significant threat.



Figure 2.2

No algorithm is completely secure from decryption, every key or cipher text can eventually be cracked using brute force, but it is the amount of time it takes to break algorithms that makes an algorithm secure. With a good brute force algorithm, it can take computers thousands to years to brute force the key. Normal encryptions like WPA/WPA2 can be broken in just 4-10 hours. WPA3 has in many ways succeeded in its task, but encryption isn't the only factor that makes a wireless network insecure.

Mistaken /Wrong Implementations

Misconfigurations or incomplete configurations are always going to be a problem, mainly because it's a human problem. A misconfiguration can range from, not enabling security rules like number of login attempts, length and complexity of credentials, not limiting the number of connections etc. Here are the biggest and most impactful examples of these.

Default Passwords

Key not being changed frequently enough

Backwards Compatibility

There are many devices that are made with backwards compatibility in mind. Why you ask? For older or different devices that don't use newer algorithms like WPA3. Thus, a lot of wireless routers will be backwards compatible with WPA2 while using WPA3 as a preferred encryption algorithm. This as one can imagine causes problems, cause the attacker can simply force the router to rely on an older method of encryption. And as we have already discussed, the older

methods are vulnerabilities which can then be misused by the attacker to gain access to the network.

Passwords not being changed frequently enough

The longer a password remains unchanged, the more likely your system is to be broken into. It is considered a good practice to change your organizational passwords in set intervals as it ensures that even if an attacker gets their hand on a password, it might be till the attacker actually gets to use that password.

No policies set to make the password complex

Simple passwords are much easier to crack with brute force or dictionary attacks. In fact, attackers create rainbow tables of hashes of common passwords to perform rainbow table attacks. It's a requirement in pretty much all organization to have complex passwords that contain multiple cases, characters, numbers etc.

People don't first realise how big this problem is, so here's a piece of stat to prove my point. The top 10 most common passwords in 2023 are;

1. 123456
2. 123456789
3. qwerty
4. password
5. 12345
6. qwerty123
7. 1q2w3e
8. 12345678
9. 111111
10. 1234567890

All of these and all other possible sequential passwords on a keyboard are present in every dictionary file use for dictionary attacks. I have no doubt that you, the reader might have used one of these passwords at some point in your life. All of humans think in the same way when it comes to creating passwords that are easy to remember. Proving the need to force people to create complex passwords. Such a problem proves to be even bigger with Wi-Fi's as it usually takes the attacker to crack of fund just one password to gain access to the entire network.

Passwords being saved in the computer or piece of paper

Another Convenience based decision that can cost you your entire network. It is strangely common in company employees to write their passwords down on a piece of paper or in a notepad file in their computer. Such file can be easily physically accessed for an insider. Thus, it is considered a bad practice to write down your passwords or to save them without any encryption.

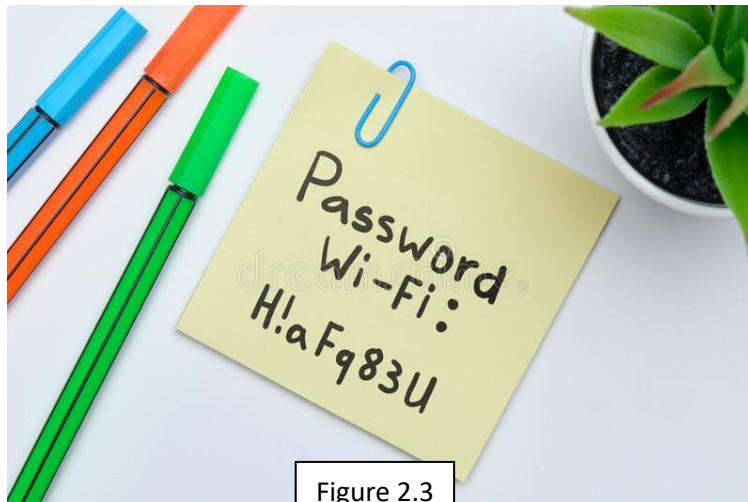


Figure 2.3

Sharing of passwords

Employees sometimes openly to share their password to “help their colleagues” or to help themselves by making someone else do something that they were supposed to do, but in Wi-Fi terms this refers to not only thoughtless sharing of passwords but also unsecure sharing of passwords. Unsupervised sharing can cause the attacker to get their hands on it and allow the attacker to observe process of sharing to capture passwords in the future.

Lack of segmentation in networks

Here segmentation refers to having different networks for different departments/sections of a company. When it comes to wireless networks, it is important to have completely different networks with different access points for different parts of a company. No segmentation can mean an attacker with access to one part of the network having access to the entire infrastructure.

Lack of internet restrictions

A company has many different types of employees and not all employees know what and what not to access through a company network. An employee might access non-trust worthy website for their own entertainment or by being fooled by a social engineer attack.

Open Access points

Open access points refer to access points that aren't secured by passwords. These are an obvious threat as an open access point means open access to anyone who can get into its range.

Human Behaviour

Lack of employee education

As discussed before, every company has different kinds of employees and not all have to knowledge of what they should or shouldn't do. Thus, employees do stupid things like click on phishing links, visit shady websites, share Wi-Fi passwords, share personal data on open networks, use any open access point their system can connect to. If the employees aren't educated then they will prove to be the biggest risk to any kind of security.

Disgruntled employee

Anger, resentment, greed or excitement. Any of these can be a reason for an employee to destroy the company he works for. Not being paid according to satisfaction, being treated wrongly, not getting a raise or promotion serve as motivating factors for this. Such an employee has internal access to your wireless networks and know about internal configurations and passwords, thus posing a huge security risk to any company/organization.

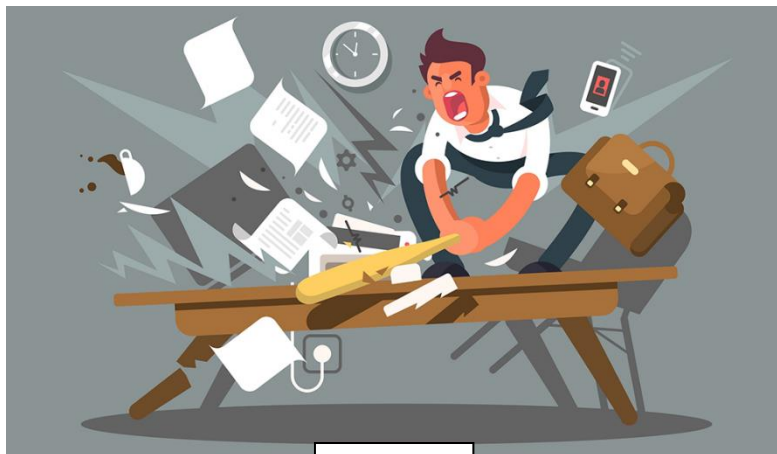


Figure 2.4

People using company network for personal purposes

This can be seen as employees being un-educated but educated employees might chose to disregard common sense justify their actions by specifying the low probability of an attack, such employees might use the company network to watch entertainment, download stuff from

illegal/shady websites, watch obscene content or do personal payment transactions. Such employees need to be controlled. Best way to do so is introduce punishments for breaking rules and letting employees understand the gravity of it all

People being allowed even after supposedly forgetting their ID cards

Employees do this all the time; they forget their access card/Id card and the security lets them and helps them pass through because of the fact that they recognise them. Now one might say this is harmless as a guard would only do this to the employees they recognise. But the fact of the matter is that employees get fired all the time, and one exception made leads to other. Next time the guard might let someone through because he received a call about them being a friend of an employee. Situations that seem okay at first get out of hand very quickly. Which is why no such exception can be allowed.

Physical Security

Lack of guards

Having a lack of/no guards is an obvious security problem as a guard can check identities and understand human behaviour. Something that machines aren't good at quite yet. Lack of guard means an attacker might just walk through the front door without anyone noticing, and once someone is inside in a big organisation no one seems to ask any questions.

Poor identification methods

Now only having a guard isn't enough. It's just as important to have good identity scanners, like ID card scanners or biometric scanners to check the identity of the employees. Its absence will make extremely easy for an attacker to physically access the company's infrastructure.

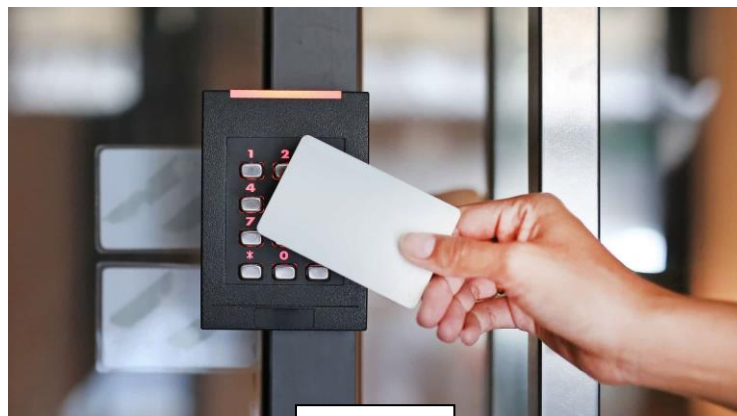


Figure 2.5

Lack of network coverage limitation

While positioning an access point for optimal coverage for the network, it is task of equal importance to make sure that the said access point isn't accessible from outside/fence the building of your organization. Lack of such limitation can allow an attacker to attack your internal network without stepping a foot inside your company's infrastructure. They can just sit by the outside wall and or the neighbouring building to carry out their attacks.

Access points being openly accessible

It's the same point we discussed earlier, and the danger of its effect amplifies when coupled with other mistakes such as lack of coverage limitation. Open access point means the attacker needs to nothing else besides simply being in the network's coverage range.

Summary

In this chapter our focus was mainly about the factors that make a wireless network insecure. We discussed the problems with encryption and authentication, about how different encryption methods were built open each other, generation by generation fixing the flaws that the last algorithm had. After that we looked at factors that stem from mistaken or wrong implementations that affect the security of a wireless network. After that we looked at the human factors that do the same and ended by discussing the bad decision in physical security that might lead to the compromise of a company/organization's wireless network security.

Chapter - 3

Wireless Threats and Attacks

Introduction

We are done looking at the workings of a wireless network and the factors that might lead to you having an insecure wireless connection. Now it's time to look at the actual threats to a wireless network and thus the corresponding attacks.

Why would an attacker attack in the first place?

Mainly there are four things that are known to motivate any cybercriminal, they are known as M.I.C.E, with the full form for each letter being;

M – Money | I – Ideology | C – Compromise | E - Ego

These factors/feelings in an individual can lead to them adopting goals like,

- Gain access to the internal network
- Steal Information for selling purposes or further attacks
- test their own skills
- To defame someone in the company or the company itself for their political or ideological stands
- To take revenge after being fired from the company
- And more

After looking at such motivations, one can easily imagine the amount of destruction an attacker can create. There are a number of possible threats that exist on wireless access points. Here are the most prominent of those threats.

Threats and Attacks on Wireless Networks

Misconfigured AP Attack

This just straight up means configuration errors by the officials in the company. This can allow any attacker to get into the network and perform attacks without raising suspicions. Key elements that play a role in this are,

- SSID broadcast

- Weak password
- Configuration error

Client Mis-Association

The attacker might setup a rogue AP outside the corporation's perimeter. The attacker first learns the SSID of the target network, then spoofs his SSID to found one and starts advertising his AP. Attacker can use this rogue AP as a channel to bypass enterprise security policies. And once a client connects to this rogue AP, we know what can happen.

Unauthorized Association

This is performed by soft APs, typically on a laptop instead of a corporate AP. Attacker might make the NIC look legitimate by running spoofing tools, he then uses the soft AP to gain access to the target network. Software APs are available on client cards or embedded WLAN radios in some PDAs and laptops; an attacker can launch these directly or through a virus programs. The attacker infects the victim's machine and activates soft APs, allowing an unauthorized connection to the enterprise network. This attacker can then steal passwords, launch attacks on a wired network or plant Trojans.

Ad-Hoc Connection Attack

Wi-Fi clients can communicate directly via an ad-hoc mode that does not require an AP to relay packets. Data can be conveniently shared among clients in ad-hoc networks, which are quite popular among Wi-Fi users. But an attacker can force a network to enable ad-hoc mode. This mode is insecure and does not provide strong authentication or encryption. Thus, such a network is readily available for an attacker to attack on.

Honeytrap AP Attack

An attacker can setup a fake hotspot that uses the same SSID as the target network. Any user that connects to different Wi-Fis will make the mistake of connecting to such a network. Such Aps mounted by attackers are called honeypots. They transmit a strong signal that overpowers the ones around and serves as an attractive option for connection.

AP MAC spoofing

An attacker can spoof the MAC address of the AP by programming a rogue AP to advertise the same identity information as that of the legitimate AP. An attacker connected to the AP as an

authorized client can have full access to the network. This type of attack succeeds when the target wireless network uses MAC filtering to authenticate clients.

Rogue AP Attack

An attacker can create a list of MAC addresses of authorized APs on the target LAN and crosscheck the list with the list of MAC addresses found by sniffing. With this into the attacker can create a rogue AP and place it near the target corporation network. Attacker then lure the target to connect to the AP. The damage to be done on success is pretty obvious, the sniffing possibilities are endless.

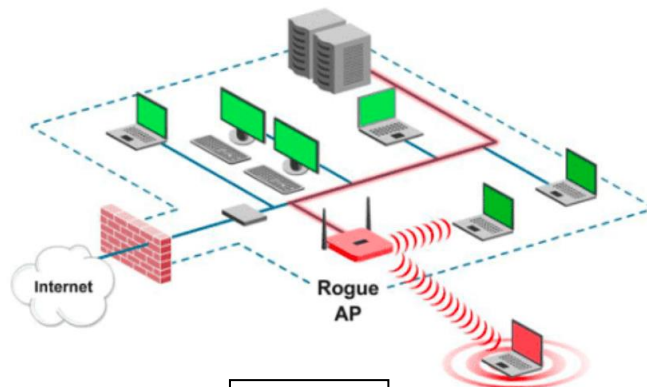


Figure 3.1

Such AP can come in different form factors,

- Compact, pocket-sized rogue AP plugged into an Ethernet port of the target network
- Rogue AP connected to corporate networks over a Wi-Fi link [will require credentials of the target network to connect]
- USB-based rogue AP plugged into a network machine [The USBs AP can share the network access of the machine it's connected to]
- Software-based rogue AP running on a network Windows machine [Attacker can set up a software-based rogue AP on the embedded/plugged Wi-Fi adapter of the target network, instead of a separate hardware device]

Such an AP can be deployed through the following steps

- Choose an appropriate location to plug in the rogue AP for maximum coverage from the connection point
- Disable SSID broadcast and any management features to avoid detection
- Place the AP behind a firewall, if possible, to avoid network scanners
- Deploy the rogue AP for a short period

Client Mis-Association

The attacker might setup a rogue AP outside the corporation's perimeter. The attacker first learns the SSID of the target network, then spoofs his SSID to found one and starts advertising his AP. Attacker can use this rogue AP as a channel to bypass enterprise security policies. And once a client connects to this rogue AP, we know what can happen

Misconfigured AP Attack



This just straight up means configuration errors by the officials in the company. This can allow any attacker to get into the network and perform attacks without raising suspicions. Key elements that play a role into this are,

- SSID broadcast
- Weak password
- Configuration error

Unauthorized Association

This is performed by soft APs, typically on a laptop instead of a corporate AP. Attacker might make the NIC look legitimate by running spoofing tools, he then uses the soft AP to gain access to the target network. Software APs are available on client cards or embedded WLAN radios in some PDAs and laptops; an attacker can launch these directly or through a virus programs. The attacker infects the victim's machine and activates soft APs, allowing an unauthorized connection to the enterprise network. This attacker can then steal passwords, launch attacks on a wired network or plant Trojans.

Ad-Hoc Connection Attack

Wi-Fi clients can communicate directly via an ad-hoc mode that does not require an AP to relay packets. Data can be conveniently shared among clients in ad-hoc networks, which are quite popular among Wi-Fi users. But an attacker can force a network to enable ad-hoc mode. This mode is insecure and does not provide strong authentication or encryption. Thus, such a network is readily available for an attacker to attack on.

Honeypot AP Attack

An attacker can setup a fake hotspot that uses the same SSID as the target network. Any user that connects to different Wi-Fis will make the mistake of connecting to such a network. Such

Aps mounted by attackers are called honeypots. They transmit a strong signal that overpowers the ones around and serves as an attractive option for connection.

AP MAC spoofing

An attacker can spoof the MAC address of the AP by programming a rogue AP to advertise the same identity information as that of the legitimate AP. An attacker connected to the AP as an authorized client can have full access to the network. This type of attack succeeds when the target wireless network uses MAC filtering to authenticate clients.

Denial-of-Service Attack

DoS refers to overwhelming the target with a large number of packets or requests to use up their processing power or bandwidth and rendering the target unusable. This is rough definition of a DoS attack. For wireless networks it we have two types of DoS attacks to look at,

- Disassociation Attack: Attacker makes the victim unavailable to other wireless devices by destroying the connectivity between the AP and client.
- De-authentication Attack: the attacker floods stations with forged de-authenticates or disassociates to disconnect users from an AP.

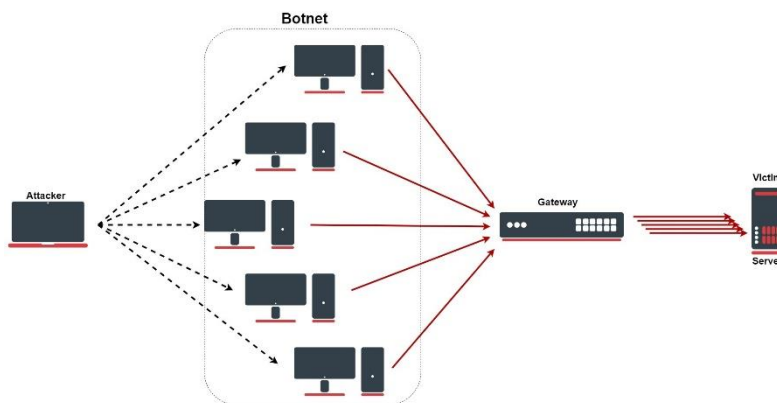


Figure 3.2

Key Reinstallation Attack (KRACK)

All secure Wi-Fi networks use the four-way handshake process to establish connections and to generate a fresh encryption key that will be used to encrypt the network traffic.

Attacker exploits the four-way handshake of the WPA2 protocol by forcing Nonce reuse. The attacker captures the victim's nonce key that is already in use to manipulate and replay cryptographic handshake messages.

Jamming Signal Attack

Overwhelming volumes of malicious traffic result in a DoS to authorized users, obstructing legitimate traffic. All wireless networks are prone to jamming, and such attacks usually blocks all communications completely. The jamming causes a DoS because 802.11 is a CSMA/CS protocol, the collision-avoidance algorithms of which require a period of silence before a radio is allowed to transmit.

aLTEr Attack

Here's a step-by-step explanation of the attack

- The attacker installs a malicious tower masquerading as a real tower
- The attacker determines the user's position and sends a packet that appears as a valid request to the real tower
- The real tower responds with the requested web link
- The attacker connects the user to unwanted or harmful websites

Wormhole Attack

The attacker locates themselves strategically in the target network to sniff and record ongoing wireless transmissions. From this location, the attacker advertises that the malicious node has the shortest route form transmitting data to other nodes in the network. To perform sniffing and to record the ongoing communication the attacker creates a tunnel to forward the data between the source and destination. The attacker sniffs the RREP message (used to discover dynamic routes) from two hosts and sends them to each other before the message reaches in a normal way, in this way the attacker fools the hosts into thinking that they have found a new shortest route. And all the traffic now passes through the attacker.

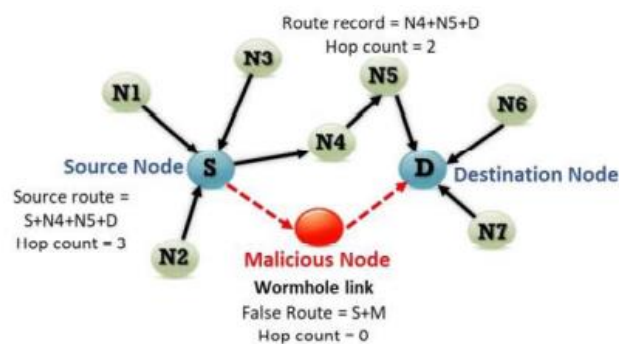
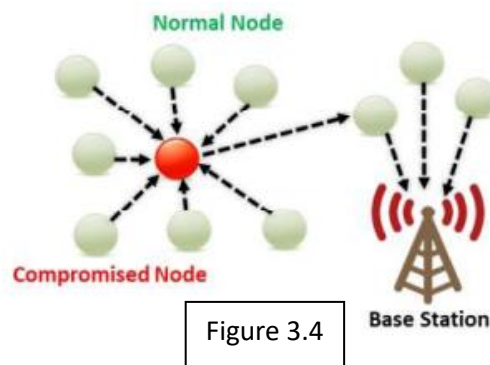


Figure 16.33: Wormhole attack

Figure 3.3

Sinkhole Attack

A variant of selective forwarding attack in which the attacker advertises a compromised or malicious node as the shortest possible route to the base station. The attacker places the malicious node near the base station and attracts all the neighbouring nodes with fake routing path information and further performs a data forging attack. Attackers use the compromised node to sniff and manipulate ongoing network transmissions. This attack can be done simultaneously with a wormhole attack, the malicious node can occupy all the network traffic and use the tunnelling technique to reach the base station faster than other nodes.



Wardriving

Wardriving involves attackers searching for wireless networks with vulnerabilities while moving around an area in a moving vehicle. They use hardware and software to discover unsecured Wi-Fi networks then gain unauthorized access to the network by cracking passwords or decrypting the router. The attacker then records vulnerable network locations on digital maps, known as access point mapping

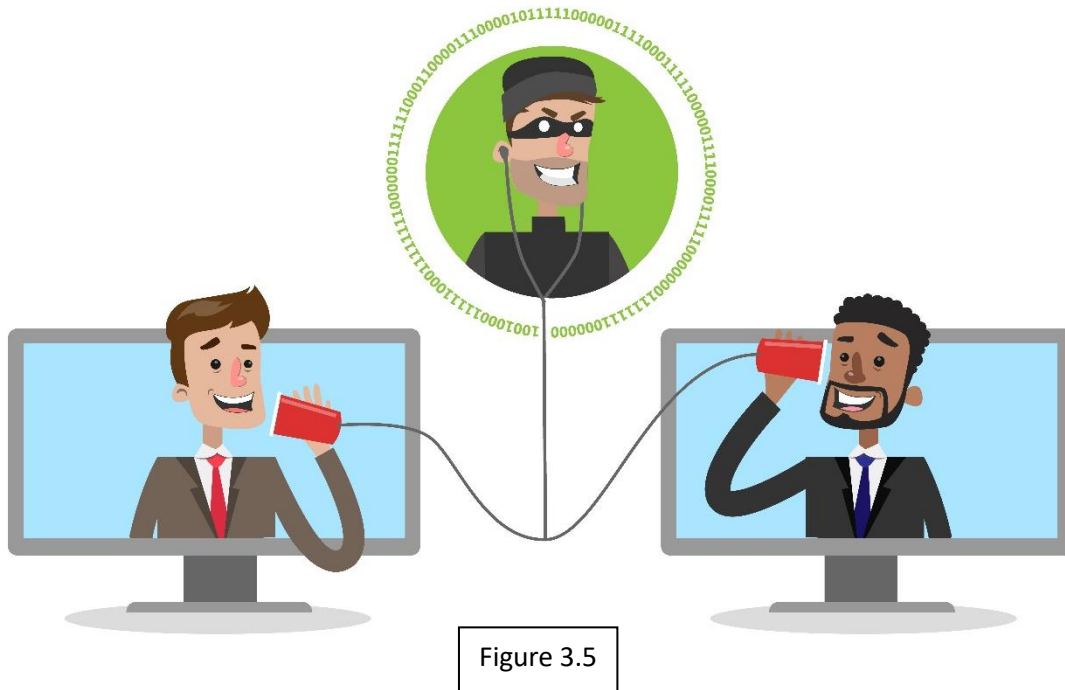
MAC spoofing

MAC spoofing simply refers to sending messages with the MAC address of another system. Spoofing as a word refers to acting as someone else or simply stealing someone's identity. Attackers can spoof wireless packets to perform attacks and blame it on the spoofed system.

Man-in-the-Middle Attack

- **Eavesdropping:** An attacker in the vicinity of a wireless network can receive radio waves on the wireless network without much effort for equipment. Multiple layers of encryption are needed to ensure that the data in such a network is secure. An attacker can crack WEP with tools freely available on the internet. Accessing email using the POP or IMAP is risky because these protocols can send an email over a wireless network without any form of extra encryption.

- **Manipulation:** Changing the captured data and sending it forward with changes is manipulation.



Man-in-the-Middle can be performed in the further steps,

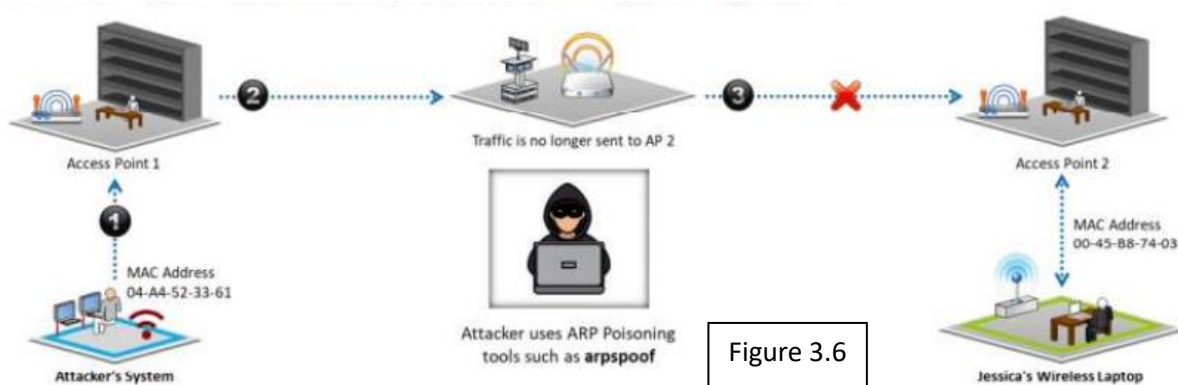
- Attacker sniffs the victim's wireless parameters
- Attacker send a DEAUTH request to the victim with a spoofed source address of the victim's AP
- On receiving the request, the victim's computer is de-authenticated and starts to search all channels for a new valid AP
- The attacker sets a forged AP on a new channel with the original MAC address (BSSID) and ESSID of the victim's AP, thereby connecting the victim to the forged AP
- After the victim's successful association to the forged AP, the attacker spoofs the victim to connect to the original AP
- The attacker positions themselves between the AP and victim, listening to all the traffic

Replay Attack

A replay attack occurs when a cybercriminal eavesdrops on a secure network communication, intercepts it, and then fraudulently delays or resends it to misdirect the receiver into doing what the hacker wants. An attacker for example, can replay a captured RADIUS packet to gain access to parts of network that they aren't supposed to have access to.

Wireless ARP Poisoning Attack

ARP poisoning exploits the lack of verification in the ARP method of finding MAC address through IP address. The ARP cache maintained by the OS is corrupted with wrong MAC address. An attacker achieves this by sending an ARP replay packet constructed with a wrong MAC address. This attack impacts all the hosts in a subnet.



Attacker first spoofs the MAC address of the victim's system and attempts to authenticate to access point 1 using an ARP poisoning tool such as arpspoof. AP1 sends the updated MAC address information to the network routers and switches, which in turn update their routing and switching tables. The traffic from the network backbone to the victim's system is sent to AP1, rather than to access point 2 (AP2).

Evil Twin

A wireless AP that pretends to be a legitimate AP by imitating its SSID. An attacker sets up a rogue AP outside the network perimeter and lures users to sign in to this AP. Tools like KARKA are used to monitor station probes to create the evil twin. This tool passively listens to wireless probe request frames and can adopt any commonly used SSID as its own SSID to lure users. Once associated the attacker may bypass enterprise security policies and gain access to network data.

Short steps of setting up of a fake hotspot (Evil Twin)

- 1) Enable Internet Connection Sharing in Windows or Internet Sharing in MAC OS X
- 2) Broadcast the Wi-Fi connection and run a sniffer program to capture passwords

Wi-jacking Attack

This attack can be used when credentials are saved in the victim's browser, when the victim accesses the same website multiple times, and when the router uses an unencrypted HTTP

connection to access the router configuration interface in the browser. Attackers can use these vulnerabilities to crack WPA/WPA2 networks without going through the handshake process.

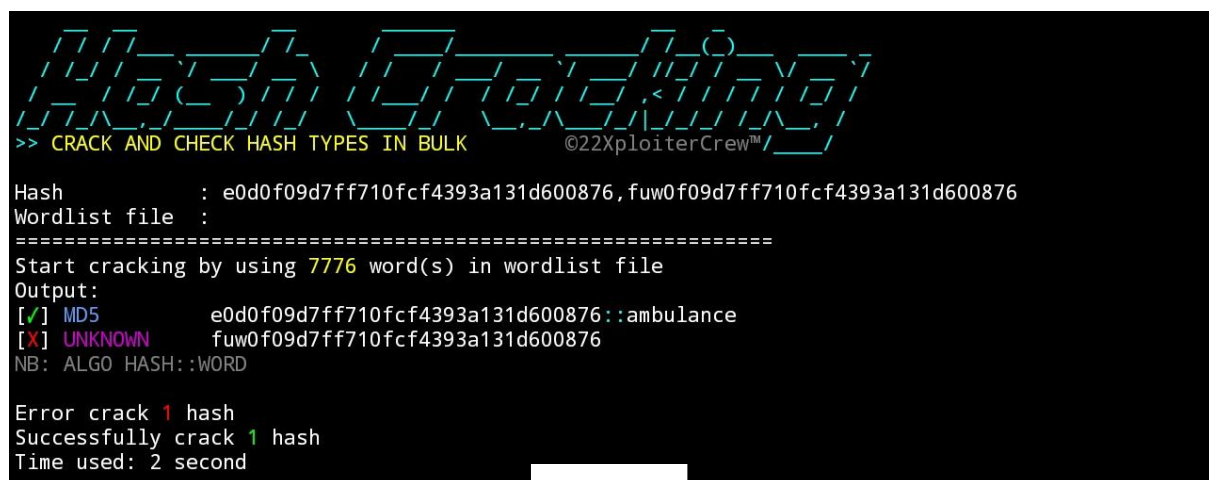
The full explanation of this attack requires an in-depth explanation that will be too much for our document here, a summarization of it involves de-authenticating the user, lure the victim to connect to malicious Wi-Fi network, use script to force a malicious website into the victim's browser. The victim will share their credentials thinking that it's the login page for the browser.

Wi-Fi Encryption Cracking Attacks

WEP Encryption Cracking

It is necessary to gather a large number of IVs which can be done simply by listening to the network traffic. An attacker can break WEP encryption through the following steps,

- Start the wireless interface in the monitor mode on the specific AP channel
- Test the capability of injection from the wireless device to the AP
- Use a tool such as aireplaying for fake authentication with the AP [With the right source MAC address]
- Start the Wi-Fi sniffing tool [capture generated IVs]
- Start a Wi-Fi packet encryption tool such as aireplay-ng in the ARP request replay mode to inject packets
- Run a cracking tool such as aircrack-ng [Using such tools the attacker can extract WEP encryption keys from the IVs]



```
>> CRACK AND CHECK HASH TYPES IN BULK ©22xploitCrew™
Hash      : e0d0f09d7ff710fcf4393a131d600876,fuw0f09d7ff710fcf4393a131d600876
Wordlist file :
=====
Start cracking by using 7776 word(s) in wordlist file
Output:
[✓] MD5      e0d0f09d7ff710fcf4393a131d600876:ambulance
[X] UNKNOWN  fuw0f09d7ff710fcf4393a131d600876
NB: ALGO HASH:WORD
Error crack 1 hash
Successfully crack 1 hash
Time used: 2 second
```

Figure 3.7

WPA/WPA2 Encryption Cracking

Following are some types of techniques used to crack WPA encryption

- **WPA PSK:** WPA PSK uses a user-defined password to initialize four-way handshake. An attacker cannot crack this, but the keys can be brute-forced using dictionary attacks.
- **Offline attack:** After being near the AP, the attacker can capture the necessary packets and crack them offline. The WPA handshake occurs over insecure channels and in plaintext, thus capturing a full handshake helps in breaking the WPA/WPA2 encryption without any packet injection.
- **De-authentication attack:** Attacker finds an actively connected client, forces him to disconnect using tools like aireplay to capture the authentication packet when the client attempts to reconnect. The attacker can crack the captured WPA key by dictionary or brute-force.
- **Brute forcing of WPA keys:** Brute force refers to trying every single possibility to break the key to ultimately decrypt the encrypted message.

WPA3 Encryption Cracking

WPA2's four-way (PSK) handshake is replaced with the Dragonfly method. Which is the strongest but still vulnerable to password-cracking attacks. Here are some methods to do so,

- 1) **Downgrade Security Attacks:** Here the attacker forces the user to follow the older WPA2 encryption method.
 - a) **Exploiting backward compatibility:** attacker installs a rogue AP supporting only WPA2, thus forcing the user to use the older ways. Which we already know how to exploit.
 - b) **Exploiting the Dragonfly handshake:** Attacker masquerades as an authentic AP. During exchange of keys the attacker informs that it does not support WPA3, thus forcing the use of the older WPA2 mechanism
- 2) **Side-Channel Attacks (Information-Leaking Attack):** During key-exchange process, the attacker launches this attack to capture leaked information. Which is further used to launch brute-force or dictionary attack. It can be implemented in the following two ways
 - a) **Timing-based attack:** Attacker analyses the time taken by the handshake to encode a certain password authentication process. The iterations of encoding process are observed and short-lists of possible passwords is made. Which obviously can then be used.

- b) **Cache-based attack:** Attacker injects a malicious JavaScript or web application in the target user's web browser. This allows a takeover of control of the user's web browser and allows the attacker to further observe memory access patterns to retrieve password information.

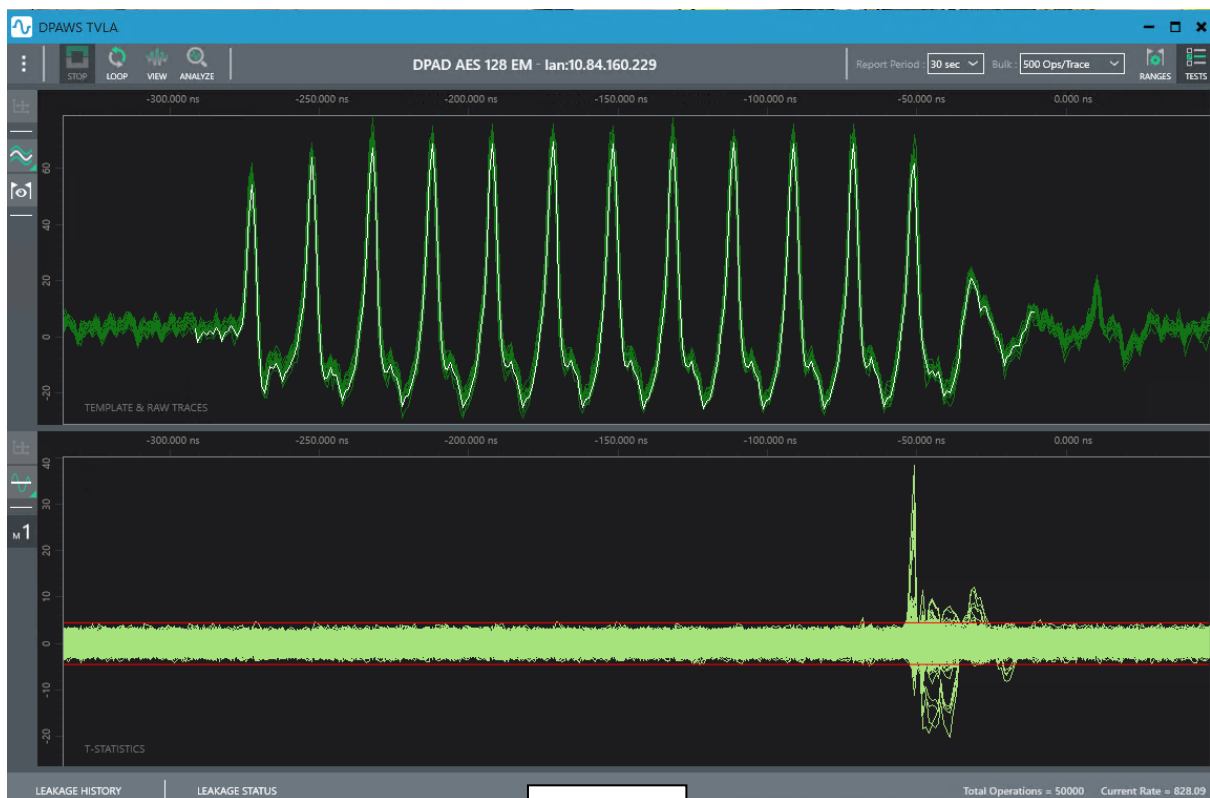


Figure 3.8

Bluetooth Hacking

Although as mentioned at the start of this documentation, our focus was going to remain on Wi-Fi based networks as they are the ones with the most variation and volatility Bluetooth still remains to be a decent attack vector. Although easier to rectify, Bluetooth attacks can still be very dangerous to any network as simply having Bluetooth enabled in your device serves as an attack vector for attackers. With that being said, here is a list and a brief explanation of attacks possible on Bluetooth devices/network.

- **Bluesmacking:** Occurs when an attacker sends an oversized ping packet to the victim's device. The attack is similar to ICMP ping-of-death attack.
- **Bluejacking:** Refers to the use of Bluetooth to send messages to users without the recipient's consent, similar to email spamming. The device initiating the connection must provide a display name which can be anything like an annoying message or advertisement. This attack does not harm, just can be annoying.

- **Bluesnarfing:** Attacker within the range of the target can use specialized software to obtain the data stored on the victim's device. Attacker exploits a vulnerability in the Object Exchange (OBEX) protocol that Bluetooth uses to exchange information. Attacker connects to the target and performs a GET operation for files with correctly guessed or known names, such as /pb.vcf for the device's phonebook or telecom/cal.vcs for the device's calendar file.
- **BlueSniff:** A proof-of-concept code for a Bluetooth wardriving utility. Useful for finding hidden and discoverable Bluetooth devices. Operates on Linux.
- **Bluebugging:** Attacker using these gains remote access to a target Bluetooth-enabled device without the victim's awareness. Attacker sniffs sensitive information and might perform malicious acts like intercepting phone calls, messages etc.
- **Blueprinting:** Foot printing technique. Attacker collects info to create infographics of the model, manufacturer etc.
- **Btlejacking:** Detrimental to Bluetooth low energy (BLE) devices. Attackers can sniff, jam and take control of the data transmission by performing MITM. Requires affordable firmware-embedded equipment and minor software coding.
- **KNOB attack:** Key Negotiation of Bluetooth (KNOB). Attacker leverages a vulnerability in Bluetooth wireless standard and eavesdrops on all the data being shared in the network such as keystrokes, chats, etc. This attack is especially detrimental to two Bluetooth-enabled devices sharing encrypted keys.
- **MAC spoofing attack:** A passive attack, attacker spoofs the MAC address of a target Bluetooth-enabled device to intercept or manipulate the data sent to the target device.
- **Man-in-the-Middle/impersonation attack:** Here the attacker captures the data meant for the target and makes sure that data doesn't reach the target directly and only reaches through the attacker's device. The attacker in this way can capture the data and edit it for their own purposes.

I'd personally recommend you learn more about these aspects if they interest. Here is a wonderful article comparing all kinds of Bluetooth attack devices and their use. To get to the article simply click on duo.com.

Besides that, [here](#) is the link to a case study on Bluetooth attacks. And a link to another wonderful article from hackernoon.com about most well common Bluetooth vulnerabilities and how to prevent the attacks that could generate from said vulnerabilities.

Real Life Wireless Hacking Instances

Taking Over an Entire City's Wireless Public Network

In 2016, an Israeli hacker successfully took over the free Wi-Fi network of an entire city. On his way home from work one day, Amihai Neiderman spotted a wireless hotspot that he hadn't seen before. It was advertised as "FREE_TLV," which is part of the citywide free Wi-Fi network set up by the local administration of Tel Aviv. Neiderman wondered how secure the network might be.



Figure 3.9

He connected to the network through one of the access points and checked what his new internet protocol (IP) address was. This is usually the public address assigned to the router through which all Wi-Fi clients access the internet. He then disconnected and scanned for open ports. He found that the device was serving a web-based login interface through port 443 (HTTPS).

After thorough analysis, he didn't find any basic vulnerabilities. He looked further. Eventually he found a **buffer overflow vulnerability** that could be exploited to take full control of the device. This case shows that skilled hackers potentially could attack thousands or tens of thousands of users by compromising large public Wi-Fi networks.

Teaching the importance of privacy

This attack happened on USA Today's technologist writer Steven Petrow. When on an American Airlines flight, he was working on an article about the dispute between the FBI and Apple, with the FBI attempting to force Apple to unlock an iPhone, and Apple refusing to do

so. Petrow was using the fee-based, in-air Gogo Wi-Fi service and had mistakenly believed he was safe.

Figure 3.10



After landing, Petrow was approached by a man in the airport terminal, who announced that he had hacked Petrow's files while on the same flight with him. The reason given by the unknown hacker was that he wanted to give Mr. Petrow the perspective of what it feels like to have personal information accessed, without giving permission. This would be one of the outcomes of story Petrow was working if Apple complied with the FBI.

Are you sure it's just a toaster?

This attack shows how simply spreading a few chips with a right plan can become a highly effective act of cyber war. In this attack devices such as toasters, irons, dashboard cameras, the small appliances, cell phones, and other electronics were found to contain a Wi-Fi chip that spread malware. Interestingly, Russian officials, who first discovered the chips in items imported from China, found the chips due to a small variation in the physical weight of the items rather than through electronic means.



Figure 3.11

Appliances and devices with these chips are able to discover and compromise unsecured Wi-Fi networks up to 200 meters away. Once a network becomes infected, hackers are able to steal information, spread malware to computers on the network, or use the network to spam or launch denial of service (DoS) attacks. All this from your toaster!

Change the default passwords!

It was found that passwords for as many as 55,000 wireless routers, provided by U.K. ISP Talk-Talk, were easily available to hackers. Most customers never change the password from the default supplied with the router. Having stolen the default passwords for routers, hackers were then able to attack customers' networks, although the hackers must be within signal range to do so, limiting to a large degree how many attacks have taken place to date.

Besides these examples it simply important to remember that cybercriminals can get access to poorly configured Wi-Fi networks – sometimes because of vulnerabilities in routers. For this reason, open and unfiltered Wi-Fi networks at an airport, coffee shop, restaurant, or hotel can put you at risk. Hackers can monitor your activity, see what sites you're visiting, and in some cases track your activity to snatch passwords to bank accounts and other important sites requiring a login.

How does a wireless attack actually happen?

A question you might have after learning about different types of wireless attacks and going through the history of well-known wireless breaches. Well, its not as easy for well thought out corporate networks but it can be very easy with normal Wi-Fi networks that we use in everyday lives. Let me prove this to you with an actual practical attack. Down below is a network diagram created in Cisco Packet Tracer. It showcases a typical wireless network. Here we have an access point, a few wireless devices connected to that access point and a server connected

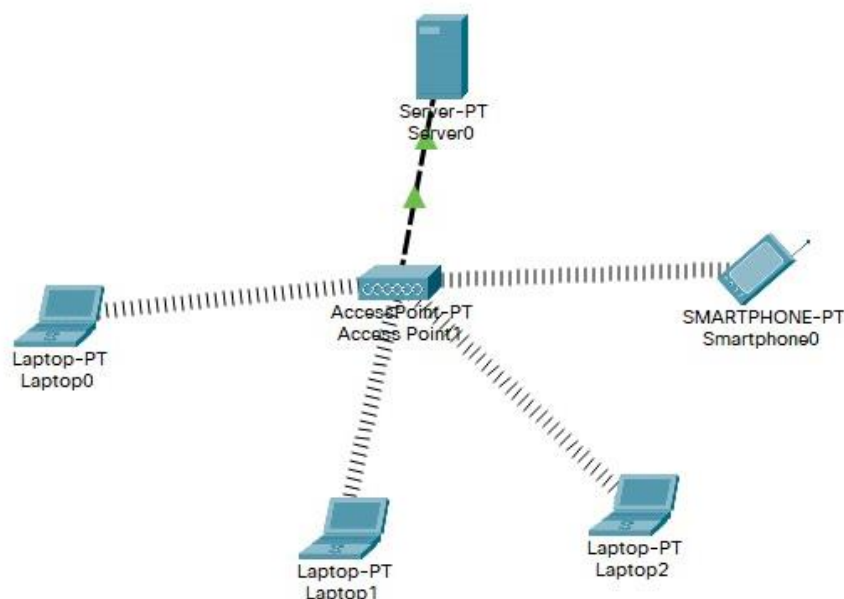
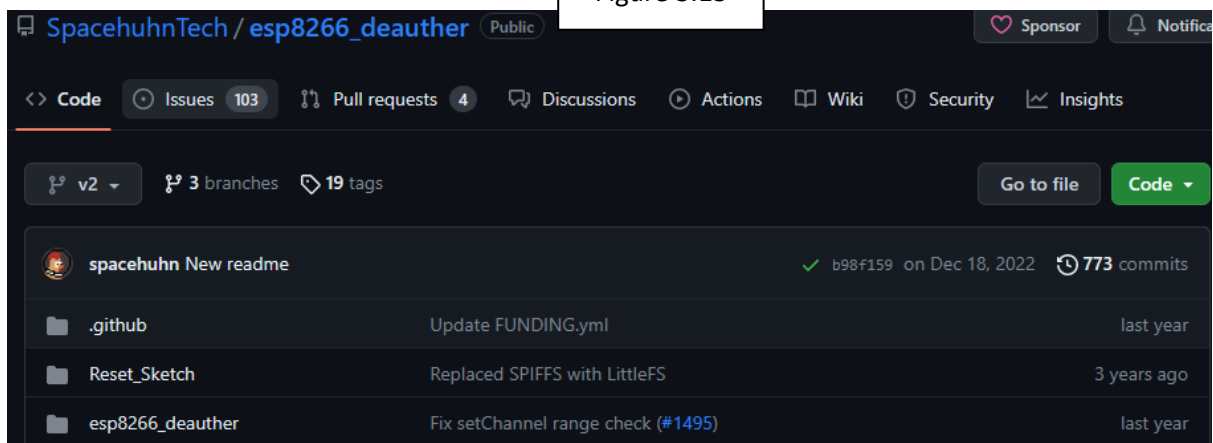


Figure 3.12

for DHCP capabilities in the network. Usually, the DHCP service is provided by the access point itself, this diagram simply aims to showcase different aspects of a wireless network.

Our aim is to do two things in this network. Do a beacon attack that creates multiple rogue access points and do a deauth attack using which all the devices currently a part of this network will be forced to disconnect from it. For this practical attack we'll use a NodeMCU ESP8266 Wi-fi serial module. Using an Arduino application, we connect to the chip using a USB cable. Arduino recognises it as a com port. Through this com port we will be able to upload the required code into the module device. The code in question was made available by user SpacehuhnTech on the GitHub repository called esp8266_deauther. [Here](#) is the link to the repository.

Figure 3.13



This is what the code in question looked like. Once the code was uploaded, I was able to directly communicate with the chip through CLI in Arduino.

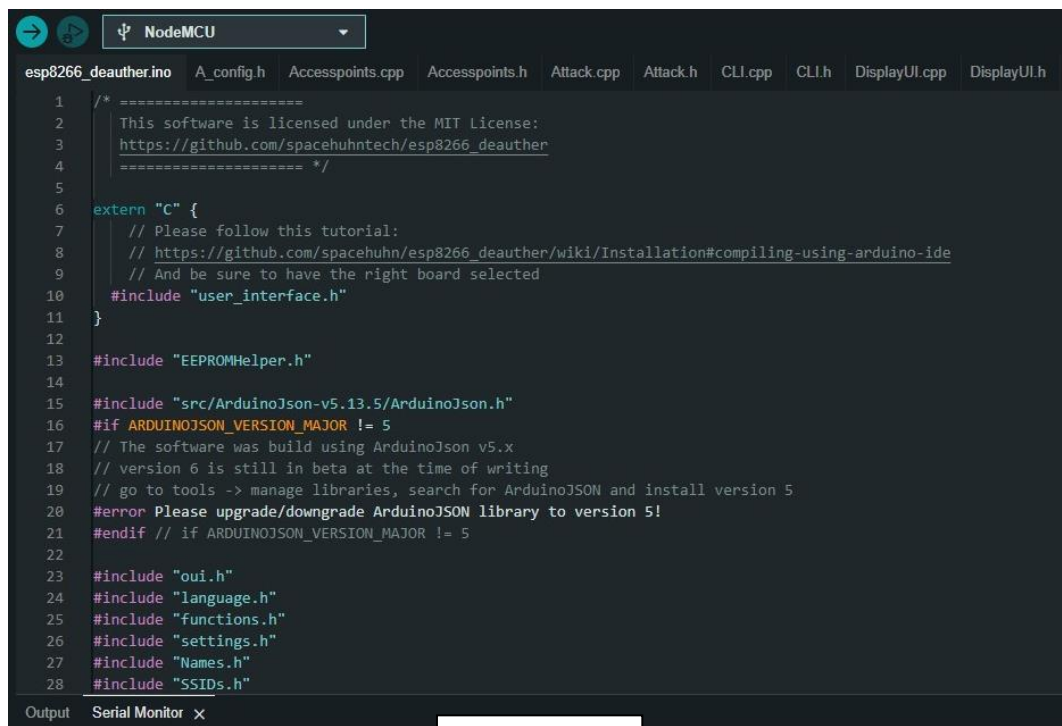


Figure 3.14

First, I scanned for available networks in the vicinity of the chip, it found two active networks. One was my own home network and another is an unknown network that most probably was from the building business we have nearby.

```

Removed all APs
Cleared station list
Starting scan for access points (Wi-Fi networks)...
[===== Access Points =====]
ID SSID Name Ch RSSI Enc. Mac Vendor Selected
=====
0 Saini house 8 -46 WPA* 24:7e:51:fe:27:42
1 Laxminarayan 4 -91 WPA2 60:a4:b7:85:f4:0c
=====

Stopped scan
Scan results saved in /scan.json
Starting Scan for stations (client devices) - 15s
Stopped Access Point
Scanning WiFi [20%]: 33 packets/s | 4 devices | 0 deauths
Scanning WiFi [40%]: 0 packets/s | 4 devices | 0 deauths
Scanning WiFi [60%]: 0 packets/s | 4 devices | 0 deauths
Scanning WiFi [80%]: 0 packets/s | 4 devices | 0 deauths
[===== Stations =====]
ID MAC Ch Name Vendor Pkts AP Last Seen Selected
=====
0 04:d6:aa:c6:52:2c 8 SamsungE 250 Saini house <1min
1 28:cd:c4:a0:f5:fd 8 31 Saini house <1min
2 00:f3:61:a0:0f:01 4 10 Laxminarayan <1min
3 6c:5c:14:44:de:41 8 Guangdon 3 Saini house <1min
=====

Started AP
Stopped scan

```

Figure 3.15

Once the right network (Saini house) was discovered, it was time to perform a beacon attack. The screenshot below shoes the command used for the attack. Right below the command it shows the status of the packets being sent for the execution of the attack. Once the attack was started, multiple new wireless networks were being shown to all the devices in vicinity of the module device.

Figure 3.16

[illegible]

In this example, the attack was clearly done as a troll of sorts. But if done by an attacker with malice in heart, this attack can be used to create multiple access points with a similar name to the original access point. Taking this attack to the next sept will be the act of creating actual rogue access points while de-authenticating users from the actual access points and tricking

them to connect to their rogue AP. Anyways, this is what one of the laptops in the area saw. Multiple APs with names inspired from the hit meme song “Never gonna give you up”.

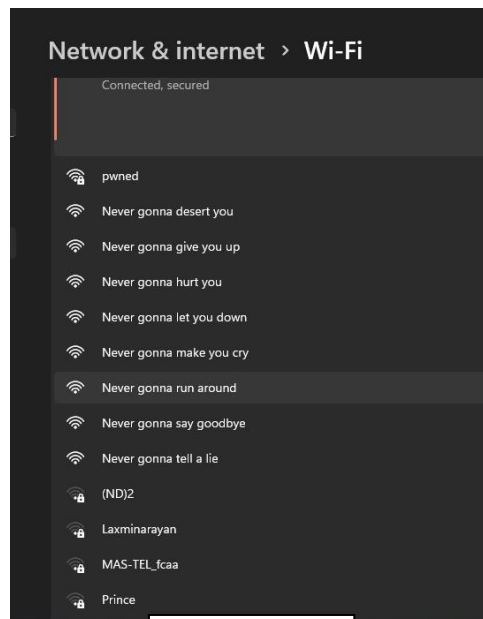


Figure 3.17

Now confusing the user by creating multiple APs will not be enough, we need to force disconnect the targets from the actual AP. So, the next attack we did is the De-auth attack. This is what the command and the attack state looked like.

```
# attack deauthall
Start attacking
[Pkt/s] All: 236 | Deauths: 0/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
[Pkt/s] All: 252 | Deauths: 252/250 | Beacons: 0/0 | Probes: 0/0
```

Figure 3.18

Once this command is run, other devices connected to the network will immediately be disconnected. The system I was connected from had lost its internet connection. And no communication could occur with other devices in the network.

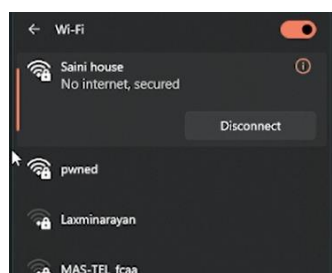


Figure 3.19

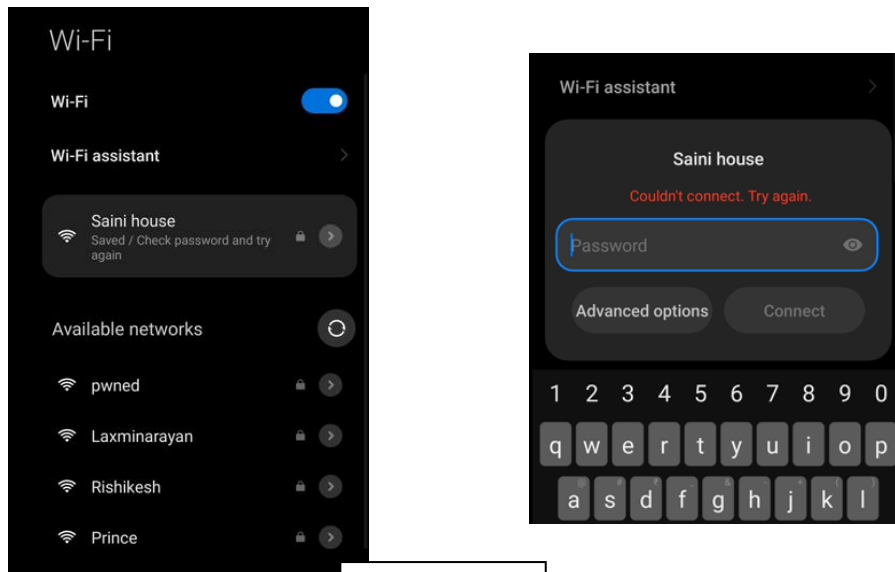


Figure 3.20

And the Mobile device connected in the network had disconnected and demanded the network credentials again. The device, after having failed so many times to connect, had assumed that the credential had been wronged or changed. Such an attack is also possible if you have an OS like Kali Linux or parrot in a system with a wireless network card. A simple command of aireplay-ng and knowledge of AP and target mac addresses (which can be acquired by simple scanning methods) will be enough. Here is an example of it.

```
root@kali:~# aireplay-ng --deauth 0 -c 98:5F:D3:4A:B1:31 -a C4:E9:84:3F:26:04 wlan0mon
21:36:31 Waiting for beacon frame (BSSID: C4:E9:84:3F:26:04) on channel 1
21:36:31 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 1|51 ACKs]
21:36:32 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 0|52 ACKs]
21:36:32 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 0|47 ACKs]
21:36:33 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [20|49 ACKs]
21:36:33 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [24|48 ACKs]
21:36:34 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 1|52 ACKs]
21:36:34 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 0|53 ACKs]
21:36:35 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 1|53 ACKs]
21:36:36 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 6|48 ACKs]
21:36:36 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [ 4|45 ACKs]
21:36:37 Sending 64 directed DeAuth. STMAC: [98:5F:D3:4A:B1:31] [28|46 ACKs]
```

Figure 3.21

Such an attack can allow an attacker to set up a rogue access point and force attackers to disconnect from the target network. The users now disconnected from the primary network would look at other networks (rogue network) with a name similar to the primary network and connect to it. Once this happens all the data of the victims of the attack would pass through the attacker's network. Through which the attacker now can gather sensitive information like passwords, credit card information etc.

Summary

This chapter was all about the dangers and possible attacks on a wireless network. We started the chapter by looking into the possible motives and reasons behind wireless attack. Mainly we tried to answer the question of Why would an attacker attack in the first place. After that we looked into all the major possible attacks that can be done on a wireless network. We then had a brief look into possible attacks on Bluetooth devices/networks too as Bluetooth has proven itself to be a good attack vector for attackers. After that we dived into a number of real-life wireless hacking instances. We discussed the way they were carried out and majorly why they were carried out. After that I got the chance to showcase and properly explain a couple of wireless attack in a real-life network. With those explanations we were able to understand how simple such attacks can be and how the attacker has so many possibilities of attacks through just a couple of attack vectors.

Chapter - 4

A Secure Wireless Network

Introduction

We have looked at how a wireless network works, what influences it, what makes it vulnerable and then we also discussed all kinds of attacks that can be performed in a wireless network. It is now time for us to understand the ways to create and keep a wireless network secure. We'll start with the most obvious part, encryption.

Layered Security

A layered approach will always be the best approach for security. For wireless networks there can be 6 individual layers that come together to contribute and create ultimate security. Sounds cheesy but it's true. The layers are as such.

- 1) **Wireless Signal Security:** A Wireless Intrusion Detection System (WIDS) analyses and monitors the RF spectrum. Activities such as increased bandwidth usage, RF interferences, and unknown rogue wireless APs might indicate a malicious intruder on the network. Continuous monitoring is the only way to detect such attacks.
- 2) **Connection security:** per frame/packet authentication provides protection against MITM attacks. Prevents sniffing.
- 3) **Device security:** Both vulnerability and patch management are important components of the security infrastructure.
- 4) **Data protection:** Encryption algorithms such as WPA3, WPA2, and AES can protect data
- 5) **Network protection:** Strong authentication ensures that only authorized users gain access to a network.
- 6) **End-user protection:** Firewalls on end users prevent attack from accessing files even if the network is breached.

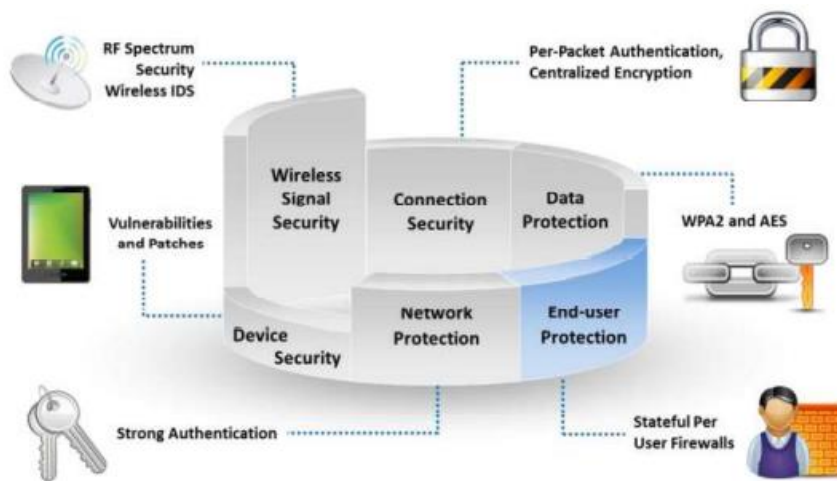


Figure 4.1

Encryption is the key

Having the right encryption is the most obvious and arguably the most important part to a wireless network's security. When it comes to Bluetooth, each version of Bluetooth came with its own default encryption algorithm. The latest version Bluetooth 5.0 uses an AES-CCM algorithm for its encryption. This encryption was made for on the fly encryption i.e., encryption that can be done in active communication. And it shouldn't come as a shock that AES-CCM is regarded as a secure encryption, well at least for now. Thus, when it comes to Bluetooth the encryption side of things are fairly simple, with each new version of Bluetooth there is a new encryption algorithm to go along with it. It is with Wi-Fi where unwanted realities can come true. There are mainly 4 different encryption algorithms that are usable with Wi-Fi which we have already discussed in depth in our past chapters. So, let's just get into what should and should not be used in a real-life scenario.

When choosing from among WEP, WPA, WPA2 and WPA3 wireless security protocols, experts agree WPA3 is best for Wi-Fi security. As the most up-to-date wireless encryption protocol, WPA3 is the most secure choice. Some wireless APs do not support WPA3, however. In that case, the next best option is WPA2, which is widely deployed in the enterprise space today.

At this point, no one should use the original wireless security protocol, WEP, or even its immediate successor, WPA, as both are outdated and make wireless networks extremely vulnerable to outside threats. Network administrators should replace any wireless AP or router that supports WEP or WPA with a newer device that's compatible with WPA2 or WPA3.



Side note: I know this image shows WPA2 as the winner, which it is when compared to WEP and WPA. But WPA3 is the undeniable winner for this day and age. So, just imagine WPA3 as the winner in the image. That will be very helpful thank you.

Backwards compatibility is dangerous

Even if you have a wireless access point with WPA3 enabled, your network might not be secure if its backwards compatible. As we had discussed in the past chapter, attackers can force a backwards compatible router to switch to a weaker encryption like WPA2 or WPA and attack that algorithm instead. Thus, it must be of upmost importance to only allow a single powerful algorithm like WPA3 to function in your network. If you have older systems in your organization that don't support higher encryption functionality then you can choose to implement network segmentation/have a different access point for those older devices or you can upgrade your devices if possible. That ultimately would be the best option.

How secure is your infrastructure physically

If you want any part of your infrastructure to be secure, physical security will always remain to be a major factor. Here are actions that you can take to physically secure your infrastructure and thus secure your networks too.

- Have security guards on every entry point of the building and make sure they maintain security logs for the entry and exit of every person from the building.
- ID card or biometrics scanners to identify employees before giving them access.
- Having metal layers in wall so that wireless signals don't travel from inside to outside and from outside to inside of the organization.
- Have security cameras with advanced capabilities like infrared vision for night time.
- Have clear rules based on who has access to which part of the organization.

- Have clear access control to different parts of your company. Every floor/segment should have a door that asks for key card or biometrics scan for access.
- Secure your network devices, keep them locked away in cabinets from physical access.
- consider hardware security measures like lockable cable ties, to secure cabling and prevent unauthorized access to ports and other interfaces.
- Educate your employees to not to share their key cards, access codes and report any suspicious activity from strangers or from the employees.
- Make sure to deploy physical alarms and sensors to make sure that sensitive parts of the infrastructure are not accessed by unwanted individuals.

Train your employees!

The human aspect of any infrastructure has the potential to be the most vulnerable one. Thus, training your employees is essential to avoid mistakes and all sorts of attacks. Here are the topics that employees need to be trained on to ensure security of your wireless network.

1. **Strong Passwords:** Employees should be taught the importance of creating strong passwords for their wireless devices and networks. They should be encouraged to use unique and complex passwords that are difficult to guess.
2. **Secrets need to be secrets:** Employees need to be taught the consequence of sharing passwords, key cards or pins with each other or strangers. In other words, something like this should not be allowed to happen as insider threats are considered to be the strongest threats to any part of an infrastructure.
3. **Encryption:** Employees should be taught about encryption and the importance of enabling it on their wireless networks and devices. They should also be reminded to use secure Wi-Fi networks when accessing sensitive data or performing work-related tasks.
4. **Network Segmentation:** Employees should be taught about network segmentation and how it can help to improve wireless security. They should be reminded to segment their wireless networks to reduce the risk of unauthorized access.
5. **Firmware Updates:** Employees should be taught about the importance of keeping their wireless devices and routers up-to-date with the latest firmware updates. These updates often contain security fixes that can help to protect against known vulnerabilities.
6. **Rogue Access Points:** Employees should be taught about rogue access points and how they can be used by attackers to gain unauthorized access to company networks. They

should be reminded to only use approved wireless networks and devices, and to report any suspicious activity to IT or security personnel.

7. **Physical Security:** Employees should be reminded about the importance of physical security when it comes to wireless devices and networks. They should be encouraged to keep their devices and routers in a secure location and to not share their wireless network passwords with anyone.
8. **Guest Networks:** Employees should be taught about the importance of guest networks and how they can help to improve wireless security. They should be reminded to set up separate guest networks for visitors and to limit their access to company data and resources.

Setup rules

Setup rules and enforce responsibilities on employees. Make sure the consequences of breaking said rules are clear, the gravity behind the enforcement of said rules is what gives said rules their purpose. Here are the rules and regulations we've been talking about.

- **Strong Password Policies:** The company should enforce strong password policies for all wireless devices and networks. Passwords should be unique, complex, and changed regularly.
- **Encryption Requirements:** The company should require encryption on all wireless networks and devices. This includes using WPA2 or higher encryption standards for Wi-Fi networks.
- **Network Segmentation:** The company should enforce network segmentation to ensure that different parts of the network are separated and secured appropriately. This includes separating guest networks from internal networks.
- **Regular Firmware Updates:** The company should enforce regular firmware updates for all wireless devices and routers. This helps to ensure that any security vulnerabilities are addressed as quickly as possible.
- **Access Control:** The company should enforce access control policies for all wireless networks and devices. This includes limiting access to authorized users only and implementing two-factor authentication where possible.
- **Monitoring and Logging:** The company should implement monitoring and logging tools to keep track of wireless network activity. This helps to identify any suspicious activity and prevent potential security breaches.

- **Incident Response Plan:** The company should have an incident response plan in place to quickly and effectively respond to any security incidents involving wireless networks or devices. This includes having procedures in place for reporting incidents, investigating them, and taking appropriate remedial action.

Here are some more specific rules that are essential for maintaining a secure wireless network.

- Sharing of passwords of wireless APs is to be done in a designated and secure manner. If possible, have a webpage-based login that requires individual credential from every employee. This would also provide access control over networks
- No outsider is to be allowed without supervision
- Company network is not to be used to access entertainment or untrusted websites
- No one should be allowed to bring their own wireless access point as it will be seen as a rogue access point.
- All access points are to be kept out of physical reach in locked cabinets
- Jamming devices are not to be allowed in the premises
- If a rogue access point is detected, it should immediately be examined and remove from the range of the premises
- Security devices like WIDS and WIPS should always remain functional and are to be updated regularly
- There should be bandwidth limits for every active connection with the wireless network

Wireless Security Devices

Wireless APs

Now access points themselves are not security devices, but having important features enabled in them goes a long way for ensuring its security. Features like;

- Stronger encryptions (WPA3)
- MAC address filtering: Helps you block suspicious MAC addresses and allow trusted ones.
- VLAN: To segment your network so that if one part of the overall network is compromised, other segments remain safe.

Wireless Intrusion Detection System (WIDS)

WIDS devices are used to monitor wireless traffic and detect unauthorized access points or clients on your network. They can also detect other security threats, such as rogue attacks, denial-of-service (DoS) attacks, and man-in-the-middle (MitM) attacks.

So, how does a WIDS detect a rogue access point? Well, it monitors the radio spectrum for the presence of unauthorized, rogue access points and the use of wireless attack tools. The system monitors the radio spectrum used by wireless LANs, and immediately alerts a systems administrator whenever a rogue access point is detected.

It is not important for these devices to be stationary. You can have handheld WIDS devices and walk around your premises to check for rogue access points.



Figure 4.3

Wireless Intrusion Prevention System (WIPS)

WIPS devices are used to prevent unauthorized access to your wireless network. They work by actively blocking unauthorized clients or access points from connecting to the network, as well as monitoring for other security threats, such as rogue attacks and DoS attacks.

A WIPS compares the MAC addresses of all wireless access points on a network against the known signatures of pre-authorized, known wireless access points and alerts an administrator when a discrepancy is found. To circumvent MAC address spoofing, some higher-end WIPS are able to analyse the unique radio frequency signatures that wireless devices generate and block unknown radio fingerprints. In addition to providing a layer of security for wireless LANs, WIPS are also useful for monitoring network performance and discovering access points with configuration errors.

There are three basic ways to deploy a WIPS.

- i. The first, primarily found at the lower-end of the market, is known as time slicing or time sharing. In this type of deployment, the wireless access point does double duty, providing network traffic with wireless connectivity while periodically scanning for rogue access points.
- ii. In the second approach, which is known as integrated WIPS, a sensor that is built into the authorized access point continually scans radio frequencies, looking for unauthorized access points.
- iii. In the third approach, which is known as WIPS overlay, sensors are deployed throughout a building to monitor radio frequencies. The sensors forward the data they collect to a centralized server for further analysis, action and log archiving. This approach is more expensive because it requires dedicated hardware, but it is also thought to be most effective.

WIPS devices are typically deployed in a rack like structure. It might be easy to mistake such racks as parts of a server. Here is what one of these devices look like.



Figure 4.4

Wireless Controllers

Wireless controllers are used to manage and monitor multiple wireless access points from a single location. They can be used to enforce security policies, configure access points, and provide centralized management and reporting capabilities.

The Wireless LAN Controller (WLC) – Lightweight Access Point (LWAP) setup is commonly utilized in the company environment to stretch an individual wireless network in a vast geographical region. This setup lets users stroll the office premise, campus, or building and still be connected to the network.

This is what one of these devices look like.



Figure 4.5

Wireless VPN Gateways

These devices are used to provide secure remote access to your wireless network. They create an encrypted tunnel between the user's device and your network, which helps to protect against eavesdropping and other security threats. Have a look at one of these beauties.

Figure 4.6



Wireless Antivirus and Firewall Appliances

These devices are used to protect your wireless network from malware, viruses, and other security threats. They provide real-time threat detection and prevention capabilities, and can be used to block malicious traffic and quarantine infected devices.

A great feature that such devices can have is packet inspection. Or in more serious cases, deep packet inspection. This refers to deep inspection of every packet that passed through this device. This lets these devices check not only the header but also the contents of a packet for potential malware. This is what a wireless firewall looks like.



Figure 4.7

Detect Rogue Access points by yourself

Now if you're running a small organization or simply can't afford to buy such expensive devices. It is possible to detect rogue access points by just using your body and your laptop. The human body attenuates Wi-Fi signal to the point where it can actually degrade your laptop's signal. Which is something that we can use to our advantage to perform this activity. So, what you need to do is;

- 1) Hold the laptop close to the body at stomach level. This will ensure that your body weakens part of the Wi-Fi signal.
- 2) Use a Wi-Fi scanner to identify the rogue APs BSSID and track its Wi-Fi signal.
- 3) Stand in one specific direction long enough to note the exact Wi-Fi signal.
- 4) From the same spot, turn 90 degrees and note the exact Wi-Fi signal. Repeat this step until you have found the direction with the strongest Wi-Fi signal.
- 5) Move 20 steps in that direction and repeat steps 3-5.
- 6) This process is repeated until you have determined the Wi-Fi signal is not getting any stronger.
- 7) Once you have found the area with the strongest Wi-Fi signal, simply look for the device! If you cannot find it in the immediate area, you might be looking for it on the wrong floor. Go to the floor above or below in the same spot, again looking for the strongest Wi-Fi signal.

Other Wireless Security Best Practices

These are some quite important but uncategorised points to follow ensuring and maintaining your own and your wireless network's security. So, here they go,

- 1) **Check your internet provider's or router manufacturer's wireless security options.**
Your internet service provider and router manufacturer may provide information or resources to assist in securing your wireless network. Check the customer support area of their websites for specific suggestions or instructions.
- 2) **Connect using a Virtual Private Network (VPN).** Many companies and organizations have a VPN. VPNs allow employees to connect securely to their network when away from the office. VPNs encrypt connections at the sending and receiving ends and keep out traffic that is not properly encrypted. If a VPN is available to you, make sure you log onto it any time you need to use a public wireless access point.

- 3) **Know exactly what you are connecting to.** If you are at an unsupervised location, make sure the find out the exact name of the access point you are supposed to connect to. Like if you're at a coffee place ask the right individuals the exact name of their hotspot or access point to prevent connecting to rogue access points. Be careful of slight variances, because attackers can use a single digit or letter difference to trick users. Next, don't browse sensitive sites, such as that of your bank, or shop online where you enter private data. Save these browsing habits for safe, private networks. If you absolutely must access a work-related site, always make sure you are using an encrypted connection.
- 4) **Don't use the same passwords that you use at other sensitive locations like bank accounts or social media accounts.** Some public Wi-Fi hotspots ask for you to login. Always ensure that the site is affiliated with the establishment and never use a password that you use with other personal shopping or banking sites. If any password is sniffed, it is only useful to the attacker if it's used somewhere else.
- 5) **Change default passwords.** Always change the default login credentials for your wireless access points and routers to a strong and unique password.
- 6) **Implement strong authentication measures such as two-factor authentication (2FA)** to ensure that only authorized users can access your wireless network. With measures like two-factor authentication and multi-factor authentication, the goal is to create multiple steps for gaining access. A hacker might be able to find out or guess your password, but having that won't mean anything if he doesn't have access to the OTPs that you receive as a second factor of authentication.
- 7) **Keep your access points and routers up-to-date with the latest firmware** updates to fix security vulnerabilities and improve performance. Firmware based vulnerabilities can be the most devastating ones as having control on any system's firmware can give the access to basically the entire system. Manufacturer's constantly send patches and updates to keep their firmware secure, such patches are to be installed immediately as hackers are always on their toes, trying to find latest vulnerabilities and exploits.
- 8) **Implement wireless intrusion detection and prevention systems (WIDS/WIPS)** to detect and prevent unauthorized access and other security threats. We have already talked on this point while discussing WIDS and WIPS devices, it was just worth to mention once again.
- 9) **Conduct regular security assessments to identify vulnerabilities and weaknesses** in your wireless network, and take steps to remediate any issues that are identified. Such

test let you know if your current configurations and practices are sufficient to ensure security. There are multiple ways of doing such test, while box, black box etc. Do your own research and figure out what kind of test is best for you and will let you find out the greatest number of vulnerabilities.

- 10) **Changing SSID or Disabling SSID** broadcast is another best practice for wireless network security. When SSID broadcast is enabled, it allows anyone within range of the wireless network to see the network's name. You can disable SSID broadcast by accessing the wireless router's configuration page and disabling the SSID broadcast feature.

The goal is to make it more difficult for unauthorized individuals to connect to the network. The SSID can still be seen if someone is within range of the network and uses a wireless network scanner, but it will not be as easily accessible.

- 11) **Disabling remote administration** is another best practice for wireless network security. When remote administration is enabled, it allows anyone with the proper credentials to access the router's configuration page and make changes to the network. This can be a security risk, as it allows unauthorized individuals to potentially gain access to the network. To disable remote administration, access the wireless router's configuration page and disable the feature. This will help to prevent unauthorized access to the network.

- 12) **Disable universal Plug and Play (UPnP)** as it is a protocol that allows devices to automatically discover and connect to each other. This can be a security risk, as it allows unauthorized devices to potentially gain access to the network. To disable UPnP, access the wireless router's configuration page and disable the feature. You can also disable UPnP on individual devices by accessing the settings menu.

- 13) **Consult the vendor about antenna positioning.** Different antennas radiate signal in different patterns. Check your vendor's documentation to verify optimal antenna positioning for your wireless network. Make sure that the antenna's coverage range and direction is utilised in the right manner and isn't pointed wrongly where it spreads its signal outside the company premises.

- 14) **Use file sharing with caution.** File sharing between devices should be disabled when not needed. You should always choose to only allow file sharing over home or work networks, never on public networks. You may want to consider creating a dedicated directory for file sharing and restrict access to all other directories. In addition, you

should password protect anything you share. Never open an entire hard drive for file sharing.

Framework

There isn't much to talk about on the framework side of things when it comes to wireless security. And it is because there just isn't enough diversity in the concept and its implementation. Other topics like website/web app security have full blown frameworks for them as those topics have incredible depth and diversity with their implementation. For wireless, things can be summarised in the best practices, like the ones that I mentioned above. Besides all of this, there is actually a framework made by NIST for wireless security. Its named **SP 800-153 - Guidelines for Securing Wireless Local Area Networks (WLANs)**. For anyone trying to build wireless network in their organisation, we recommend this. I can unironically say, in NIST we trust.

Summary

In this chapter we first looked into the layers of security that one can setup to secure their wireless network. After that we discussed the importance of encryption, mainly the importance of having good encryption. After that we understood why backwards compatibility is a dangerous feature, something that can negate having a good encryption. Then we looked into the different actions you can take to physically secure your organisation with the context of wireless network security. We then understood what to exactly train your employees about to ensure the security of your wireless network. Then we looked into the organizational rules that one must have to achieve this goal of security. We then looked at the use and function of all kinds of different wireless security devices. After all of this we discussed the best practices for ensuring and maintaining wireless security. Now a lot of the best practices had already been talked about earlier in the unit, these practices we're the unstructured and standalone points that are short and simple to understand but carry a huge importance. And finally, we ended the unit and with it this documentation by talking about Framework for Wireless Network Security.

Conclusion

Modern Wireless communication has been in use for more than 20 years now, and we have settled ourselves well enough with it. Its convenience has obviously created a noticeable attack surface, but our experience with this technology has enabled us to anticipate pretty much all of the threats that stem from it.

These threats have been studied by many, which has led the world to understand and integrate the required solutions in their products and implementations. Most of the risks now stem from wrong implementations, convenience and laziness-based practices, lack of knowledge and in some cases lack of funds. Companies now let their customers know what the right configurations and implementations are. Best practices are readily available. And wireless security devices are smart enough to identify risks in a respectfully efficient manner. Overall, we all are in a good position where the responsibilities for all the sides are clear. All we need to do now is to understand our individual/collective responsibilities and act on them.

As a final summary, in this document we were able to discuss;

- The inner workings of a wireless network
- Factors that influence a wireless network
- Possible threats and vulnerabilities in a wireless network
- possible attacks on a wireless network
- And finally, the ways, methods and tools required to secure a wireless network.

I'd like to personally thank you for going through all of my work. And wish you the best for your journey in the world of IT.

References

- Emerson.com
- Techtarget.com
- Webopedia.com
- Fortinet.com
- w7cloud.com
- Youtube.com
- stratixsystems.com
- community.cisco.com
- CEHv11 - Certified Ethical Hack V11
- cybernews.com
- pluralsight.com
- dot11ap.wordpress.com
- nibusinessinfo.co.uk
- openai.com
- juniper.net
- tutorialspoint.com
- digitalairwireless.com
- NIST - National Institute of Standards and Technology

Image References

- dreamstime.com
- getsafeandsound.com
- Recruiter.com
- vaisala.com
- researchgate.net
- istockphoto.com
- semiengineering.com
- cisco.com
- sonicguard.com